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Neoproterozoic tectonic structure of the Yenisei Ridge and formation of the western margin of the Siberian craton based on new geological, paleomagnetic, and geochronological data

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Abstract

There are continuing issues concerning the formation and reconstruction of the geographic position of the Neoproterozoic Yenisei Ridge—a key element of the western framing of the Siberian craton and the Central Asian orogenic belt. This study focuses on the inner structure, composition, and boundaries of the Central Angara terrane, which is the largest in the Transangarian segment of the Yenisei Ridge. We propose a scheme of fault deformation of the region and demonstrate that the fault tectonics of the Central Angara terrane is distinct from that of adjacent terranes. We study in detail the Yeruda pluton granitoids of the Teya complex, which indicate accretionary-collisional magmatic events in this terrane prior to its collision with Siberia. New geochemistry and SHRIMP U–Th–Pb zircon geochronology of the granites indicate that they formed at 880–860 Ma in a collisional setting. Integrated petromagnetic and paleomagnetic investigations yield a paleomagnetic pole that is significantly different from the corresponding Neoproterozoic interval of the apparent polar wander path (APWP) for Siberia. The difference in paleolatitudes between the Central Angara terrane and the Siberian craton at the time of the Teya granites formation was at least 8.6 degrees, which equals a latitudinal separation of at least 1000 km. We consider various possible positions for the terrane relative to the Siberian craton. These results demonstrate that the 880–860 Ma magmatic events in the Central Angara terrane are not related to events in the western margin of the Siberian craton. Therefore, they do not indicate the existence of a Grenville-age orogenic belt in this location, as proposed by some authors.

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Introduction

The Yenisei Ridge is a late Precambrian orogenic belt located along the Yenisei River and is a key structural feature of the western framing of the Siberian craton and Central Asian fold-and-thrust belt. Therefore, deciphering its tectonic structure and formation conditions is necessary for regional and global geodynamic reconstructions of the Eurasian lithosphere (Didenko et al., 1994; Dobretsov, 2003; Dobretsov and Zonenshain, 1985; Dobretsov et al., 1995, 2003; Gordienko, 2006; Khain, 2001; Khain et al., 1997; Kheraskova et al., 2010; Li et al., 2008; Metelkin et al., 2007; 2012; Pisarevsky et al., 2008; Sengor et al., 1993, 2014; Vernikovsky et al., 2003a, 2004; Zonenshain et al., 1990).

The Yenisei Ridge is a large accretionary-collisional structure, well defined by geological and geophysical data from the Siberian platform in the east and the West Siberian plate in the west (Belyaev and Basharin, 2001; Gliko, 1957; Kirichenko, 1963; Krylov et al., 1967; Staroseltsev et al., 2003; Surkov et al., 1996; Vernikovsky et al., 2009). In previous publications (Vernikovsky and Vernikovskaya, 2006;

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Vernikovsky et al., 1999, 2003) we demonstrated that several terranes of different geodynamic origin could be identified within the orogen (Fig. 1). In the Transangarian segment, located north of the Angara Fault and Angara River, the terranes are composed of mainly Neoproterozoic rocks—the Central Angara granite-metamorphic terrane, the East Angara passive continental margin terrane, and the Isakovka island arc terrane.

This study is primarily concerned with the Central Angara granite-metamorphic terrane. Based on our previous analysis of the geology, particularly the granitoid magmatism, of the Central Angara terrane, we proposed that this terrane formed far from the Yenisei Ridge (Vernikovsky and Vernikovskaya, 2006; Vernikovsky et al., 2003). However, the solution of this problem required paleomagnetic data that can demonstrate the kinematics of this terrane in the Neoproterozoic. Resolving this issue is essential to achieving accurate geodynamic reconstructions for the Siberian craton and its Meso-Neoproterozoic framework. It will also let us determine whether the western Siberian margin was involved in Grenville-age collisional events related to the formation of the Rodinia supercontinent (Daziel, 1991; Li et al., 2008; Metelkin et al., 2005; Pisarevsky et al., 2008; Powel and Meert, 2001).

In this contribution, we present new detailed structural, petrological, geochemical, geochronological, and paleomagnetic data for the Yeruda pluton granites (Teya complex) of the Central Angara terrane. We provide a general geological-geophysical analysis of the terrane's structure and its Neoproterozoic kinematics, and demonstrate the Central Angara terrane's allochthonous origin.

Tectonics and inner structure of the Central Angara terrane

The Central Angara terrane is 350–400 km long and 50–80 km wide, and occupies the central region between the Isakovka island arc terrane that was thrust onto the Central Angara terrane from the west and the East Angara terrane in the east, which represents a fragment of the Meso-Neoproterozoic passive margin of the Siberian paleocontinent (Fig. 1). The boundaries of the Central Angara terrane are the largest fault zones in the region—Priyenisei and Tatarka–Ishimba—mainly of upthrow-thrust kinematics (Vernikovsky et al., 2003). These are suture zones, characterized by positive magnetic and gravity anomalies, which reflect a high proportion of mafic and ultramafic igneous rocks (Karataev, 1961; Staroseltsev et al., 2003; Surkov et al., 1996).

Along the Priyenisei fault zone are fragments of ophiolites and island arcs of the Priyenisei (Isakovka) belt (Khain et al., 1997; Vernikovsky et al., 1993; Volobuev, 1993; Zonenshain et al., 1990). Zircon U–Pb crystallization ages have been published for plagiogranites—697 ± 4 Ma (Vernikovsky et al., 2001) and pegmatoid gabbroic rocks—682 ± 4 Ma (Kuzmichev et al., 2008). At the boundary with the East Angara terrane, along the Ishimba thrust, are tectonic lenses and sheets of ophiolites of the Ribny–Panimba complex that crop out clearly on the north side of the Angara River (Chernykh, 2000; Kheraskova, 1999; Matushkin, 2010; Postelnikov and Kheraskova, 1997; Vernikovsky et al., 2000). The age of these ophiolites, estimated from $^{40}Ar/^{39}Ar$ data for amphibole and plagioclase from the metagabbro and diabase is 1050–900 Ma (Chernykh, 2000; Vernikovsky et al., 2000), which probably reflects the time of their accretion to the Central Angara terrane. Similar $^{40}Ar/^{39}Ar$ dates for amphibole from the same rocks confirm a late Mesoproterozoic age for this tectonothermal event: 1043 ± 12 Ma (Nozhkin et al., 2008a), 1222 ± 14 Ma (Matushkin, 2010), 1183 ± 10 Ma, 1087 ± 11 Ma (Likhanov and Reverdatto, 2015).

The Central Angara terrane is mostly composed of metamorphosed Precambrian terrigenous, terrigenous-carbonate, and carbonate deposits, which are traditionally included in the Teya (with the Garevka sequence), Sukhoy Pit and Tungusik series (Brovkov et al., 1985; Kachevsky et al., 1998; Kirichenko, 1963). The lower part of the Teya series is dominated by highly argillaceous kyanite-(andalusite ± sillimanite)-staurolite-garnet and biotite-sillimanite gneisses, overlain by alternating biotite-amphibole schists, quartzites, amphibolites, and marbles. The Sukhoy Pit series is mainly metamorphosed terrigenous strata—metasandstones with interbeds of polymictic conglomerates and gravelstones, silty metasandstones, and phyllites that range from green to dark gray and black and only in the upper parts of the series are overlain by terrigenous-carbonate deposits. The overlying Tungusik series is an alternation of carbonaceous and argillaceous shales, overlain by limestone and dolomite. The compositions of all series correspond to passive continental margins deposits. Regional metamorphism for the lower parts of the section reaches amphibolite and epidote-amphibolite facies (mostly in the Teya series), whereas most of the Sukhoy Pit and Tungusik series does not exceed greenschist facies (Bovin, 1982; Brovkov et al., 1985; Kozlov and Lepezin, 1995; Likhanov and Reverdatto, 2014; Vernikovsky et al., 2003).

The relationships between these series, and their ages have been highly debated due to attempts to correlate them with deposits similar in composition in so-called different structural-formational zones (Khomentovsky, 2007; Khomichev, 1999; Postelnikov, 1980) that were in fact terranes of differing geodynamic origins and paleogeographic positions (Vernikovsky et al., 2009). Another type of mistake was the tendency of researchers to "build up the thickness of the lower part of the section" and consequently the search for Archean and early Proterozoic rocks especially in highly metamorphosed areas (Kachevsky et al., 1994, 1998).

Metamorphic rocks of the Central Angara terrane were intruded by precisely dated Neoproterozoic granitoids of the Teya, Ayakhta, Glushikha, and Tatarka complexes (Fig. 1, 2), which permits us to determine the ages of the host sedimentary-metamorphic series (Romanova et al., 2012; Vernikovsky and Vernikovskaya, 2006; Vernikovsky et al., 2003, 2007, 2008). The only firm constraint is that the host rocks are older than the oldest intruding granites at 880 Ma. Therefore, we provisionally consider the metamorphosed Teya (including the Garevka sequence) and Sukhoy Pit series to be Mesoproterozoic in age. There is no evidence for Paleoproterozoic or

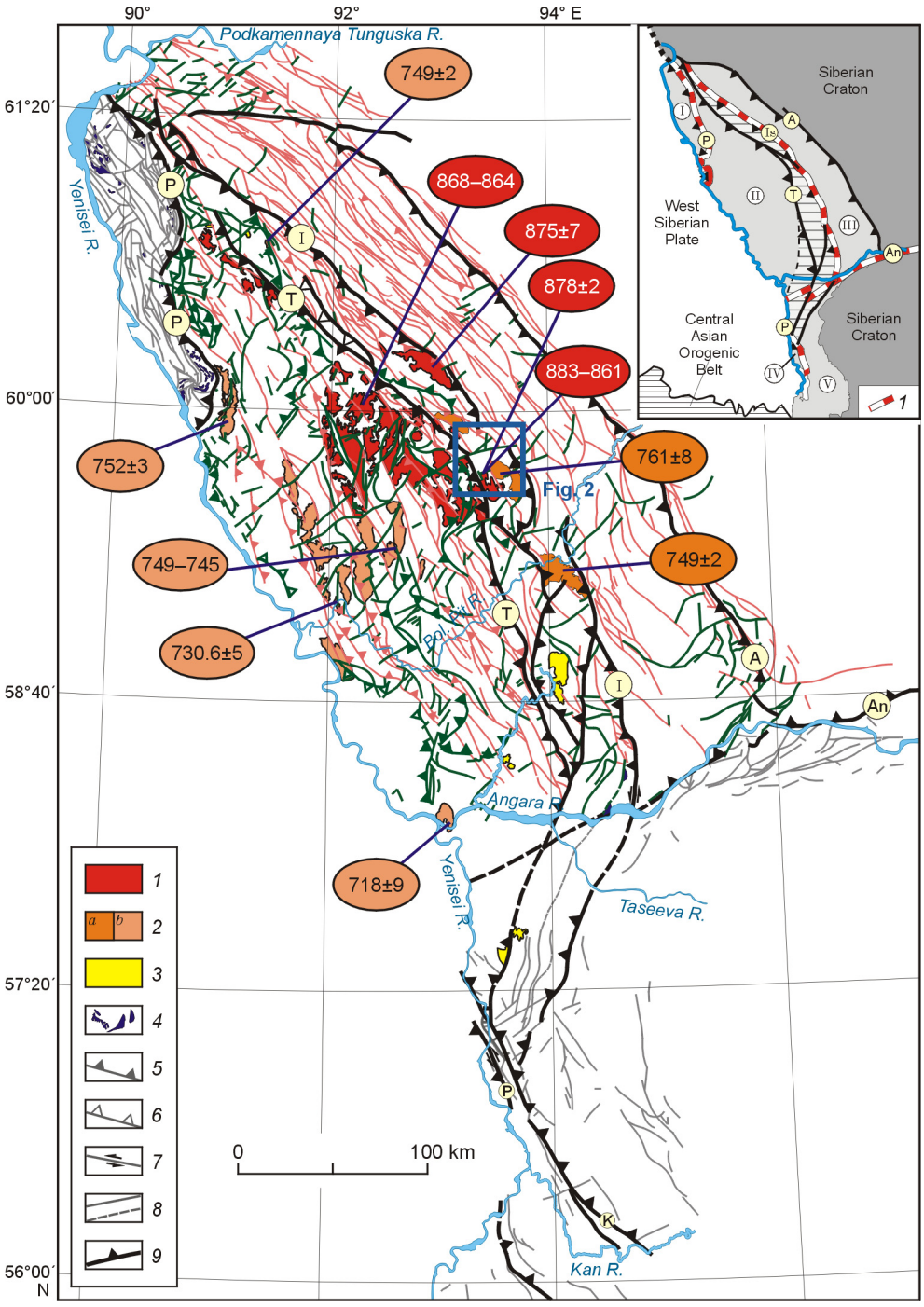


Fig. 1. Scheme of fault tectonics and granitoid magmatism in the Transangarian segment of the Yenisei Ridge, based on Kachevsky et al. (1998), Vernikovsky et al. (2003), Vernikovsky et al. (2009). *1*, Teya complex collisional granitoids (883–861 Ma); *2*, Ayakhta complex syncollisional granites and leucogranites (761–749 Ma) (*a*), Glushikha complex postcollisional leucogranites (752–718 Ma) (*b*); *3*, Tatarka active margin complex: alkaline and nepheline syenites, ijolites, urtites, trachytes and carbonatites (711–630 Ma); *4*, ophiolites of the Isakovka (NP_3) and Rybny–Panimba (MP) belts; *5*, upthrows and thrusts; *6*, normal faults; *7*, established sinistral strike-slip component in faults; *8*, identified and inferred faults; *9*, largest faults and sutures; Colored ovals contain U–Pb zircon ages of granitoids in Ma (from Vernikovskaya et al., 2002, 2003, 2004, 2006; 2007; Vernikovsky et al., 2002, 2003, 2007; Postnikov et al., 2005; and this study). Fault zones: An, Angara; P, Priyenisei; I, Ishimba; T, Tatarka; A, Ankinov; K, Kovdor. Inset: Is, Isakovka terrane; II, Central Angara terrane; III, East Angara terrane; IV, Predivinsk terrane; V, Angara Kan terrane. Dashed field, Tatarka–Ishimba suture zone. *1*, terrane boundaries.

Archean rocks within the sedimentary-metamorphic sequences of the Transangarian segment. The deposits of the Teya, Sukhoy Pit, and Tungusik series are unconformably overlain by essentially unmetamorphosed late Neoproterozoic to Lower Cambrian terrigenous and terrigenous-carbonate deposits of the Chingasan, Vorogovka, and Chapa series and their stratigraphic analogues (Khomentovsky et al., 1972; Khomichev, 1999).

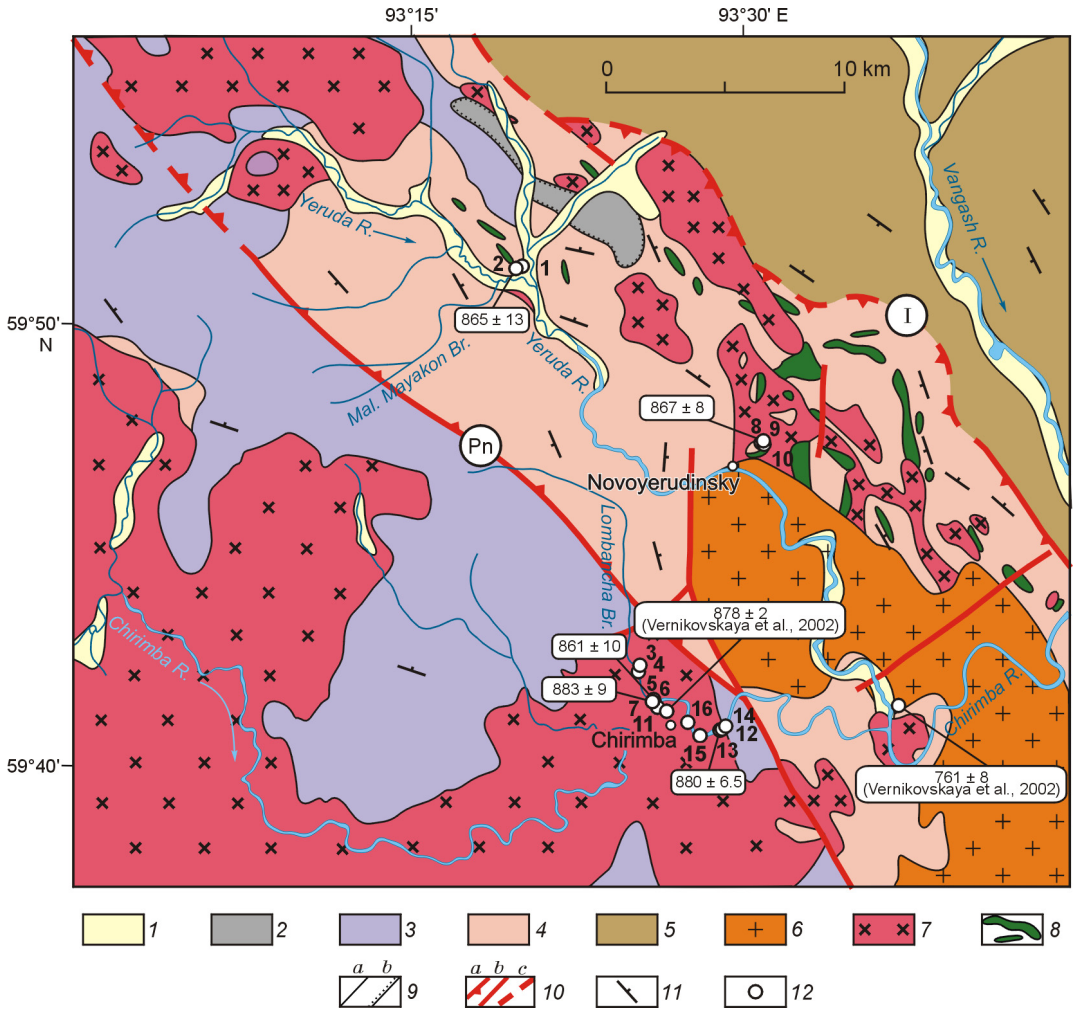


Fig. 2. Geological map of the region between the Chirimba and Yeruda Rivers (modified after Kachevsky et al., 1998; Vernikovskaya et al., 2002). *1*, Quaternary system; *2*, red sandstones, conglomerates (Taseeva series, NP_3); *3–4*, Meso-Neoproterozoic metamorphosed deposits: *3*, carbonaceous shales, siltstones and sandstones, limestones (Sukhoy Pit series); *4*, marbles, quartzites, gneisses and crystalline schists, amphibolites (Teya series); *5*, weakly metamorphosed sedimentary deposits of the passive continental margin of the Siberian craton; *6*, Chirimba pluton granitoids, Ayakhta complex (761 Ma); *7*, Yeruda pluton granodiorites, granites, plagiogranites, quartz diorites, Teya complex (883–861 Ma); *8*, amphibolitized gabbros (Indygla complex), NP_2(?); 9, stratigraphic and intrusive contacts (*a*), unconformity in stratified sequences (*b*); *10*, upthrows and thrusts (*a*), other faults (*b*), inferred faults (*c*); *11*, bedding attitude; *12*, sampling sites (Table 1). Numbers in white boxes show SHRIMP U-Pb zircons ages (in Ma) (this study). Large faults: I, Ishimba suture; Pn, Panimba thrust.

Geophysical investigations indicate that the tectonic block between the Priyenisei and Ishimba sutures has a mushroom-shaped and imbricate morphology (flower structure), which among other attributes is due to the fan-like orientation of the large longitudinal fault planes (Staroseltsev et al., 2003). The sutures that bond the terrane can be traced to depths of at least 15–25 km, where they flatten out and are truncated by the younger suture between the western margin of the Yenisei Ridge and the basement of the West Siberian plate. Aside from the particularities of the tectonic structure of the Central Angara terrane's boundaries, its deformations character is different to that of the adjacent terranes (Fig. 1). For example, deformation of the East Angara terrane, which represents a dislocated area of Siberian platform cover (the Outer Yenisei Ridge), corresponds completely to lateral stress from the west manifest as an imbricate combination of thrusts in which linear folds of NNW strike are associated with co-directional thrust faults with west-dipping fault planes. At the same time, the intensity of fold and disjunctive tectonics decreases from the NW to the SE. This agrees with the identified decrease in the ages of tectonomagmatic events in this direction (Fig. 1) and can be considered as one of the indicators of oblique collision between the Central Angara terrane and Siberia.

The structure of the Central Angara terrane itself is significantly more complex, with multiple transverse NE striking faults as well as folds (ref. Fig. 1). Additionally this terrane is deformed by uplift along east and northeast dipping thrust faults, which reflects the mushroom-like morphology described above. Multiple lateral normal faults of both NE and SW dip (Kachevsky et al., 1998) are typically bent and apparently deformed by secondary events. Detailed structural investigations including microstructural analyses in the middle and southern parts of the Tatarka–Ishimba suture zone (Matushkin, 2010) indicate two deformation stages of roughly

east–west lateral stress, one of which had a sinistral strike-slip component. Sinistral shear deformation confirms the oblique character of the collision, whereas the mushroom-like shape of the orogen suggests a transpressional setting.

Specific regularities are noted as well in the location of granitoid plutons of the Central Angara terrane. The oldest Transangarian granitoids allocated to the Teya complex (Fig. 1) are clearly concentrated in the northern half of the terrane and lack any consistent orientation (Komov, 1969). In contrast, the younger Neoproterozoic granitoid complexes are typically located along northwest-striking suture zones and faults and are elongated in the same direction. In addition, the Teya complex plutons often commonly show evidence of complex and possibly multistage folding, including the bending of pluton contacts together with folding of the host rocks (Kachevsky et al., 1998). These plutons are also significantly dislocated by faults of both groups (NW and NE strike). U–Pb ages of the 878–857 Ma for Teya, Kalama, and Yeruda plutons from this complex (Nozhkin et al., 1999; Vernikovskaya et al., 2002; Vernikovsky et al., 2007), are markedly older than the age of the Ayakhta (syncollisional), Glushikha (postcollisional) complexes that formed 761–749 and 752–718 Ma, respectively (Vernikovskaya et al., 2002, 2003, 2006, 2007; Vernikovsky and Vernikovskaya, 2006).

In summary, geophysical data, the occurrence of ophiolite complexes, and the structural control on magmatism, all indicate that the boundary between the Central Angara terrane and the margin of the Siberian craton is a major terrane suture.

Yeruda pluton granites

Granites of the Yeruda pluton were selected for detailed petrological, geochemical, geochronological, and paleomagnetic studies because previous investigations showed that they formed about 100–120 Ma before the emplacement of the syncollisional Ayakhta and postcollisional Glushikha granites, which means probably outside of the Siberian craton. Preliminary rock-magnetic investigations of these granites also indicated that they are good targets for paleomagnetic study.

Geological position. The Yeruda pluton is located in the eastern Central Angara terrane in the middle of the Tatarka–Ishimba suture zone (Fig. 1, 2). The main studies of the granites were performed in the central part of this pluton (Fig. 2), in outcrops along the Chirimba River, 8–12 km upstream of where it is joined by the Yeruda River. Studies were also conducted on smaller granite plutons that outcrop in the northeast bank of the Yeruda River close to Novoyerudinsky town and in the upper reaches of the Yeruda River near the junction of the Malyi Mayakon brook. The morphology of the plutons, their significant area, and the undulating form of their contacts, indicate shallow dipping and large volume.

The host rocks epidote-amphibolite and greenschist facies metasedimentary rocks of the Sukhoy Pit and Teya series, deformed in an imbricate thrust structure, in which the opposite dipping Ishimba suture and Panimba fault play leading roles. Deformation related to the Panimba fault include ubiquitous jointing that dips moderately to the northeast. Within the tectonic zone between the Panimba thrust and the Ishimba suture, on the northeast bank of the Yeruda River (Fig. 2, sampling site 2) in a road exposure are sheared carbonaceous, quartz-sericite and quartz-biotite-sericite schists with graphite and garnet that contain lenses of sheared metagabbroic rocks. The attitude of the succession is parallel to the schistosity: subvertical with a northeast or southwest strike. In such subvertical zones of viscoplastic strain, various microstructures and kinematic indicators (schistosity deformation, delta-structures, pressure shadows around garnet porphyroblasts, grain rotation, and boudinage of layers) indicate a vertical, thrust-uplift shear component. Deformation corresponds to lateral strain in the upthrown side of the Panimba thrust, indicating tectonic transport from northeast to southwest.

The metagabbroic rocks mentioned above form multiple small subtabular bodies, deformed conformably with the host rocks (Kornev et al., 1974) and have been assigned to the Indygla magmatic complex by survey geologists. They can be traced to the northwest and southeast along the Ishimba suture for tens of kilometers and occur exclusively within rocks of the Teya series. Their geological position indicates that they are older than both the folding and the granitoid magmatism of the Teya complex.

Petrography and geochemistry. The Yeruda pluton is dominated by medium-grained amphibole-biotite granodiorite and biotite granite with porphyry-like feldspar inclusions (up to 1 cm). There is a smaller amount of biotite leucogranite and fine-grained, plagioclase-porphyritic granite, as well as dikes of plagiogranite with aplitic textures. In addition to plagioclase and quartz, the groundmass contains small amounts of potassium feldspar, whose concentration in dikes does not exceed 10%. Moreover, in the contact zone of the pluton (near Novoyerudinsky town) there are dikes of fine-grained plagiogranite, and quartz diorite that intrude metagabbroic rocks. All studied igneous rocks have a variably developed gneissic texture, defined by elongated melanocratic minerals. In the granite these minerals include biotite (up to 10 wt.%), whereas the granodiorite also contains relict amphibole grains. Secondary minerals include biotite, muscovite and chlorite. In some areas, the rocks are carbonatized.

Determination of main, rare earth and other trace elements in whole-rock samples was performed using the quadrupolar Agilent7500ce (Japan) and the high-resolution Element2 (Germany) mass spectrometers following the methodology published in Smirnova et al. (2010) in the laboratories of the Institute of the Earth Crust SB RAS (Irkutsk). The relative error for the x-ray fluorescence method was 1–5% and for the ICP-MS analyses was 5–10%.

In granites and granodiorites the SiO_2 content varies from 65 to 74 wt.%, the alkalis sum—from 5.44 to 7.78 (Table 1). According to the classification by (Frost et al., 2001) their composition corresponds to the calc-alkalic magmatic series and the marginal areas with the calcic and alkali-calcic series (Fig. 3). They are characterized by wide variations in the

Table 1. Chemical composition of igneous rocks in the region of the Chirimba and Yeruda rivers interfluve

Component	A-08-6	A-08-7-4	A-10-4-1	A-08-7-1	A-10-6	A-08-6-1	A-10-4-3	A-10-3
	3	7	12	5	16	4	14	11
SiO_2, wt.%	65.00	67.39	67.83	70.01	72.35	72.48	72.56	72.79
TiO_2	0.68	0.59	0.45	0.40	0.26	0.28	0.30	0.28
Al_2O_3	16.02	14.73	15.80	13.88	14.28	14.06	14.27	14.03
Fe_2O_3*	5.86	5.44	4.27	3.41	2.40	2.47	2.90	2.60
MnO	0.16	0.09	0.04	0.06	0.04	0.06	0.04	0.05
MgO	1.83	1.43	1.07	0.96	0.48	0.69	0.46	0.92
CaO	4.03	1.68	2.54	2.68	1.64	1.71	1.95	0.26
N_2O	3.59	4.26	3.86	3.38	3.41	3.31	3.46	5.04
K_2O	1.85	2.33	3.45	3.85	4.20	4.22	3.73	2.73
P_2O_5	0.16	0.18	0.21	0.11	0.08	0.09	0.08	0.08
LOI	0.86	1.82	0.50	1.19	0.88	0.56	0.26	1.25
Total	100.01	99.94	100.03	99.92	100.02	99.93	100.03	100.03
Ga, ppm	23	21	–	21	–	18.20	18	–
Rb	123	129	–	210	–	194	143	–
Sr	329	351	–	341	–	236	182	–
Y	29	34	–	19.50	–	15.40	14.50	–
Zr	233	477	–	262	–	129	255	–
Nb	15.10	16.90	–	15.20	–	10.90	13.40	–
Sn	1.82	<0.5	–	1.02	–	0.57	1.85	–
Cs	3.80	1.82	–	0.99	–	4.60	–	–
Ba	320	834	–	914	–	708	812	–
La	46	82	–	55	–	37	47	–
Ce	87	154	–	96	–	68	91	–
Pr	9.20	17.20	–	11	–	7.40	9.40	–
Nd	34	59	–	36	–	27	36	–
Sm	6.10	8.40	–	4.80	–	4.40	6.10	–
Eu	1.41	1.69	–	2.10	–	0.99	0.62	–
Gd	5.10	5.80	–	3.20	–	3.20	4.92	–
Tb	0.82	0.75	–	0.44	–	0.38	0.48	–
Dy	4.80	3.60	–	2	–	2.30	3.41	–
Ho	0.89	0.70	–	0.38	–	0.45	0.66	–
Er	2.60	1.78	–	1.15	–	1.27	1.96	–
Tm	0.43	0.26	–	0.18	–	0.19	0.26	–
Yb	2.60	1.65	–	1.28	–	1.33	1.72	–
Lu	0.39	0.27	–	0.20	–	0.19	0.26	–
Hf	4.60	9.60	–	5.80	–	2.80	7.40	–
Ta	0.77	0.63	–	0.98	–	0.91	–	–
Pb	11.6	6.50	–	35	–	38	–	–
Th	12	24	–	13.50	–	14.20	27	–
U	2.40	1.91	–	2	–	1.64	2.33	–
$N_2O + K_2O$	5.44	6.59	7.31	7.23	7.61	7.53	7.19	7.77
Na_2O/K_2O	1.94	1.82	1.12	0.88	0.81	0.78	0.93	1.84
$(La/Yb)_N$	12.01	33.76	–	29.19	–	19.18	18.56	–
Eu/Eu*	0.77	0.74	–	1.63	–	0.80	0.43	–
Sr/Y	11.34	10.32	–	17.48	–	15.32	12.55	–

Note. Sequence sample numbers (1–16) correspond to the numbers of sampling sites on Fig. 2.
A-08-6, A-08-7-4, A-10-4-1, A-08-7-1, A-10-6, A-08-6-1, A-10-4-3, A-10-3, A-10-4-2, A-08-7-3, and A-10-5, Granodiorites, granites, and leucogranites; A-08-2

Table 1 (*continued*)

Component	A-10-4-2	A-08-7-3	A-10-5	A-08-2	A-08-1	A-08-8	A-08-8-1	A-08-8-2
	13	6	15	2	1	8	9	10
SiO_2, wt.%	73.02	73.58	74.03	74.08	74.52	49.74	64.98	70.54
TiO_2	0.29	0.29	0.15	0.18	0.15	1.99	0.78	0.33
Al_2O_3	14.02	13.58	13.82	14.86	15.01	12.93	15.84	15.29
Fe_2O_3*	2.71	2.60	1.82	1.49	1.24	15.07	5.58	2.58
MnO	0.04	0.04	0.03	0.02	0.03	0.20	0.06	0.03
MgO	0.44	0.67	0.43	0.48	0.35	5.84	1.70	0.92
CaO	1.56	0.71	0.89	1.22	1.18	9.16	4.83	3.71
N=O	3.75	5.14	3.23	4.31	4.04	3.29	3.82	5.00
K_2O	3.44	2.47	5.12	2.35	2.48	0.60	0.64	0.41
P_2O_5	0.28	0.07	0.05	0.06	0.04	0.18	0.18	0.15
LOI	0.45	0.80	0.45	0.86	0.90	0.86	1.57	0.97
Total	100.00	99.95	100.03	99.9	99.92	100.11	99.98	99.92
Ga, ppm	18	15.60	–	21	19.50	17.70	23	20
Rb	132	81	–	68	90	27	44	24
Sr	148	225	–	784	797	402	531	318
Y	10	14.20	–	5.50	8.50	38	25	9.80
Zr	237	182	–	83	108	105	388	325
Nb	13.20	9.70	–	2.10	2.20	11.40	14.50	12.90
Sn	3.13	3.20	–	<0.5	36	<0.5	<0.5	1.25
Cs	–	1.29	–	0.75	1.29	0.84	1.06	7.40
Ba	647	584	–	646	730	80	282	236
La	31	16.60	–	8.10	5.40	10.60	23	24
Ce	70	39	–	16.80	21	26	66	91
Pr	6.70	4	–	1.99	1.29	4.20	5.50	6.30
Nd	26	13.60	–	8	5.10	19.20	20	21
Sm	4.61	2.80	–	1.63	1.35	5.20	4.50	4.10
Eu	0.59	0.55	–	0.49	0.34	1.91	1.62	0.71
Gd	3.84	2.10	–	1.24	0.90	5.80	3.90	3
Tb	0.47	0.31	–	0.16	0.14	1.01	0.50	0.44
Dy	2.62	1.72	–	0.87	0.79	6.30	2.60	2.20
Ho	0.46	0.32	–	0.15	0.14	1.34	0.44	0.37
Er	1.41	0.83	–	0.40	0.42	3.70	1.21	0.99
Tm	0.18	0.13	–	0.06	0.07	0.57	0.18	0.16
Yb	1.29	0.89	–	0.34	0.42	3.70	1.08	1
Lu	0.19	0.13	–	0.058	0.06	0.54	0.16	0.15
Hf	7	4	–	2.20	2.90	2.60	7.40	8.50
Ta	–	0.84	–	0.21	0.21	0.70	0.90	1.21
Pb	–	13.30	–	24	24	0.07	10.3	21
Th	25	14.60	–	2.90	2.60	1.23	11.9	33
U	2.87	2.70	–	1.09	1.44	0.27	1.29	3
$N_2O + K_2O$	7.19	7.61	8.35	6.66	6.52	3.89	4.46	5.41
Na_2O/K_2O	1.09	2.08	0.63	1.84	1.63	5.51	5.95	12.22
$(La/Yb)_N$	16.32	12.67	–	16.8	8.73	1.95	14.47	16.30
Eu/Eu*	0.26	0.69	–	1.05	0.94	1.06	1.18	0.62
Sr/Y	14.80	15.85	–	142.55	93.76	10.58	21.24	32.45

and A-08-1, plagiogranites (C-type adakites); A-08-8, A-08-8-1, and A-08-8-2, quartz diorite and plagiogranite dikes and host metagabbro. Element contents in ratios are normalized to chondrite (McDonough and Sun 1995): $(La/Yb)_{CN}$ and $Eu/Eu^* = Eu_{CN}/[Gd_{CN} \times Sm_{CN}]^{0.5}$. Dash, Not analyzed.

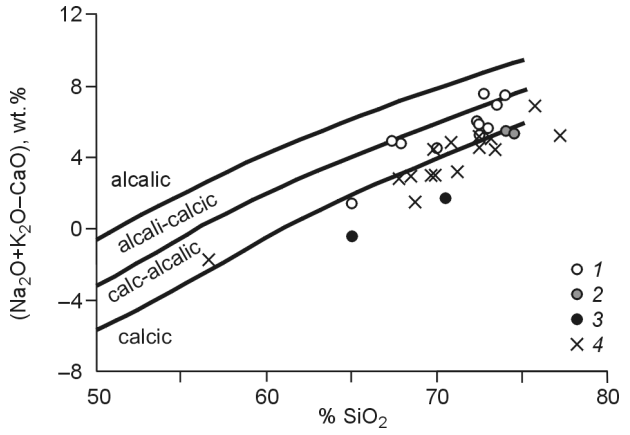


Fig. 3. $Na_2O + K_2O$–CaO vs SiO_2 diagram (MALI alkali-calc index) from (Frost et al., 2001) for igneous rocks of the Yeruda pluton in the region between the Chirimba and Yeruda Rivers. *1*, granodiorites, granites, leucogranites; *2*, plagiogranites (C-type adakites); *3*, plagiogranites dikes and associating quartz diorites; *4*, Teya complex rocks from (Vernikovsky et al., 2007). This diagram and Fig. 4 use data from Table 1.

Na_2O/K_2O ratio (from 0.78 to 2.08) and moderate to high content of K_2O (1.85–4.22). One sample of a more leucocratic rock (SiO_2 = 74.03 wt.%) the alkali sum increases to 8.35 wt.%, and the Na_2O/K_2O ratio decreases to 0.63. The A/CNK index in the rocks varies from weakly metaluminous to peraluminous composition from 0.95 to 1.2 mol.%.

The aplitic plagiogranite dikes (Table 1) are characterized by more elevated SiO_2 contents (74.08–74.52 wt.%), decreased alkali sum (6.66–6.52), predominance of Na_2O over K_2O (the Na_2O/K_2O ratio varies between 1.63 to 1.83) and low K_2O content (about 2.4 wt.%). These rocks are peraluminous (the A/CNK index varies from 1.25 to 1.31) and their data occupy a marginal area between the calcic and the calc-alkalic magmatic series (Fig. 3).

The plagiogranites and quartz diorites that intrude metagabbros (Table 1) have even lower alkali sums (4.46–5.41 wt.%) and elevated Na_2O/K_2O ratios (5.95–12.22). They have tholeiitic compositions (K_2O content does not exceed 0.18) and correspond to the calcic magmatic series (Fig. 3). They have low A/CNK indexes (0.57–1.01) that correspond to metaluminous and subaluminous rock variations.

Trace element distributions of the granites and granodiorites (Fig. 4*a*, *b*) have relatively flat HREE distributions, moderate $(La/Yb)_{CN}$, ratios varying from 12.01 to 33.76 and Sr/Y ratios (13.32–17.48). They have both positive and negative Eu anomalies (Eu* = 0.43–1.63) and negative Ba, Ta, Nb, Sr, P and Ti anomalies. The aplitic plagiogranite dikes (Fig. 4*c*, *d*) are characterized by positive Sr anomalies, negative Ta, Nb, P and Ti anomalies and a lack of Eu anomalies. They have high Sr/Y ratios in the interval of 142.55–93.76, demonstrating the sharpest REE distribution angle among all studied rocks. These rocks display characteristics of C-type adakites according to (Xiao and Clemens, 2007). The plagiogranite and quartz diorite dikes associated with metagabbros have contents of Ta, Nb, Sr, P, Nd and Sm that are close to those of the metagabbros (Table 1, Fig. 4*e*, *f*), with positive Th, U, and Ce anomalies as well as negative and small positive Eu anomalies. $(La/Yb)_{CN}$ ratios are low and vary in the interval 14.47–16.30. Sr/Y ratios do not exceed 32.4.

Based on our analysis of the geochemical data for the groundmass of the Yeruda pluton granitoids, taking into account data published previously for other plutons of the Teya complex (Vernikovskaya et al., 2002; Vernikovsky et al., 2007) we consider them to be products of continental crustal magmatic sources formed in collisional settings. The aplitic plagiogranite dikes in the Yeruda pluton display C-type adakite characteristics. These are potassium enriched adakites, which are probably products of magma formed in the base of the continental crust due to extreme thickening at the final stages of a collision process. Such rocks have been identified in China and, according to experimental data (Xiao and Clemens, 2007), they could have formed during the high-temperature melting of rocks similar to the tonalite-trondhjemite-granodiorite association. The geochemical particularities of the diorite-plagiogranite association, like those of the host metagabbros (tholeiitic, mostly metaluminous composition, relatively flat HREE distributions), indicate the presence of a (depleted) mantle component in the source, which probably means they are of island arc origin.

Geochronology. U–Th–Pb measurements of zircons from the Yeruda pluton samples (A-08-7-3, A-08-8-1, A-08-2 and A-08-7-4) were conducted using the SHRIMP II ion microprobes at the John de Laeter Centre of Mass Spectrometry at Curtin University (Perth, Australia), following analytical procedures described by Wingate and Kirkland (2014). Zircons from four samples, together with zircon standards, were mounted in an epoxy disc, which was polished to section the crystals to half their thicknesses, and photographed in transmitted and reflected light. Following application of a 40 nm coating of high-purity gold, the zircons were imaged using cathodoluminescence (CL) and backscattered electron (BSE) techniques.

U–Th–Pb ratios were determined relative to the Temora standard zircon (416.8 Ma (Black et al., 2004)). Uranium and thorium concentrations were determined relative to the 91500 standard (81 ppm ^{238}U (Weidenbeck et al., 2004)). Data were reduced using Squid 2.50 (Ludwig, 2009) and Isoplot 3.7 (Ludwig, 2003) software, using decay constants recommended by Steiger and Jäger (1977). Correction for initial or common Pb employed regression from initial Pb, which assumes that measured Pb is a mixture of initial and radiogenic Pb, and that the Pb–Pb and U–Pb isotopic systems are concordant. Analyses of minerals of the same age can be expected to plot (by amounts proportional to their common-Pb contents) along a mixing line between initial $^{207}Pb/^{206}Pb$ (calculated according to the model of Stacey and Kramers (1975)) and radiogenic Pb, on concordia. Data falling significantly to the left of the mixing line suggest the presence of xenocrystic material, whereas dispersion to the right may indicate ancient or recent radiogenic-Pb loss. The lower intercept of the mixing line with concordia indicates the average age of the sample. Note that because the data are all close to concordia, the method of common-Pb correction makes no significant difference to

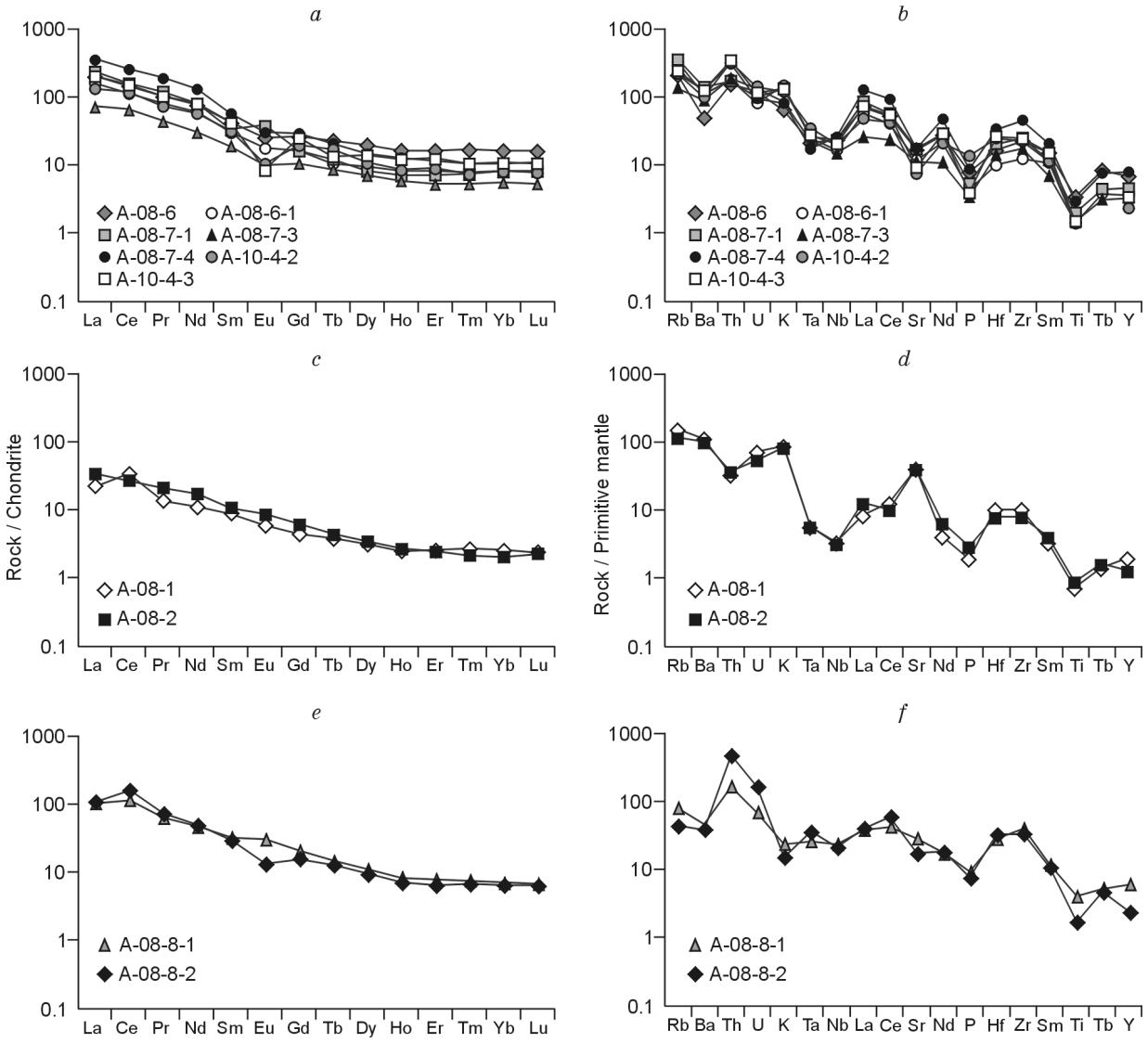


Fig. 4. REE distributions (*a*, *c*, *e*) and spider-diagrams (*b*, *d*, *f*) for igneous rocks of the Yeruda pluton in the region between the Chirimba and Yeruda Rivers. (*a*) and (*b*), Granodiorites, granites, leucogranites; (*c*) and (*d*), plagiogranite dikes (C-type adakites); (*e*) and (*f*), plagiogranite dikes and associating quartz diorites. Chondrite and primitive mantle contents are taken from (McDonough and Sun, 1995).

calculated ages. Mean ages are quoted below with 95% confidence intervals ($t\sigma\sqrt{\text{MSWD}}$) and including $^{238}U/^{206}Pb$ reproducibility and calibration uncertainties).

Zircons from granodiorite sample A-08-7-4 are colorless and subhedral to euhedral. The crystals are up to 250 μm long and equant to elongate, with aspect ratios up to 4:1. Most exhibit concentric zoning (Fig. 5*a*). Eleven analyses of 11 zircons are concordant to slightly discordant, and ten are tightly grouped (Fig. 6*a*). One analysis yields a $^{238}U/^{206}Pb^*$ date of 326 Ma suggesting that this zircon is a contaminant introduced during sample processing. The remaining ten analyses indicate mainly moderate to high uranium and thorium contents (Table 2). A regression from initial Pb ($^{207}Pb/^{206}Pb = 0.8991$ at 883 Ma) through uncorrected data yields an intercept of 883 ± 9 Ma (MSWD = 1.3) for eight analyses. Two analyses indicate younger dates (822, 782 Ma) that may reflect loss of radiogenic Pb. The date of 883 ± 9 Ma is interpreted as the crystallization age of the sample.

Zircons from quartz diorite sample A-08-8-1 are mainly colorless and subhedral to euhedral. The crystals are up to 250 μm long and equant to elongate, with aspect ratios up to 8:1. Concentric zoning is common (Fig. 5*b*). Ten analyses of ten zircons are concordant to slightly discordant (Fig. 6*b*), and indicate mainly moderate to high uranium and thorium contents (Table 2). A regression from initial Pb ($^{207}Pb/^{206}Pb = 0.8978$ at 867 Ma) through uncorrected data yields an intercept at 867 ± 8 Ma (MSWD = 1.9), interpreted as the crystallization age of the sample.

Zircons from plagiogranite sample A-08-2 are similar to those described above in color and shape. The crystals are up to 300 μm long and mainly elongate, with aspect ratios up to 6:1. Most exhibit concentric zoning and some crystals contain metamict zones that are dark in CL images (Fig. 5*c*). Some

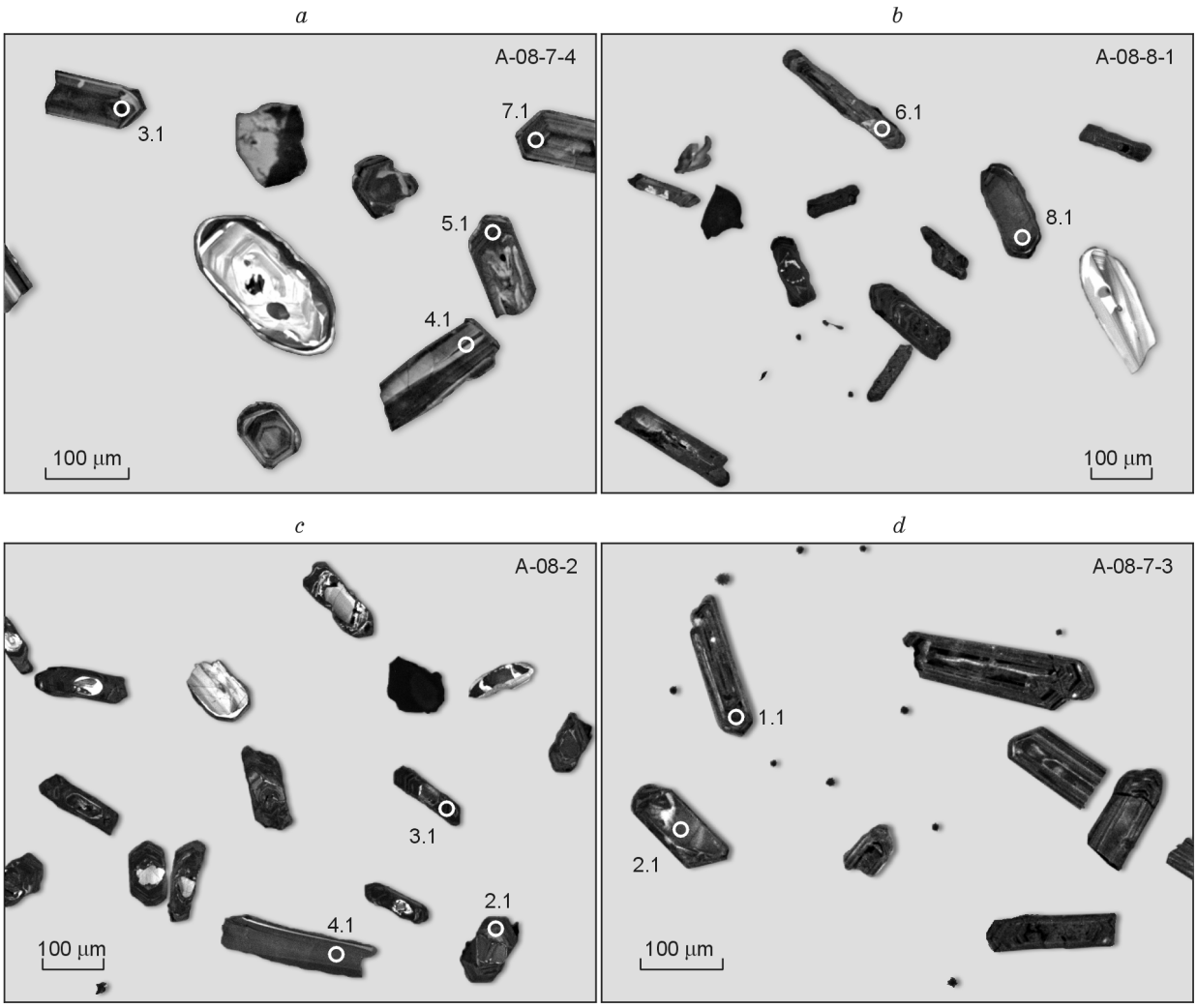


Fig. 5. Cathodoluminescence (CL) microphotographs for representative zircons from igneous rocks of the Yeruda pluton. Numbered circles indicate approximate location of analytical sites.

zircons appear to contain older cores. Fourteen analyses were obtained from 13 zircons (Table 2). Three analyses >10% discordant, and two that indicate very high variation in isotope ratios, are excluded from further consideration. The remaining nine analyses are concordant to slightly discordant (Fig. 6*c*). A regression from initial Pb ($^{207}Pb/^{206}Pb = 0.8977$ at 865 Ma) through uncorrected data indicates a concordia intercept at 865 ± 13 Ma (MSWD = 1.1) for four analyses. Three analyses yield significantly older dates (c. 1777, 1383, 913 Ma) that are interpreted as the ages of inherited zircons. Two high-U analyses are significantly younger (818, 790 Ma) and interpreted to reflect loss of radiogenic Pb. The date of 865 ± 13 Ma is taken to be the best estimate of the crystallization age of the sample.

Zircons from leucogranites sample A-08-7-3 are colorless to dark brown and mainly euhedral. The crystals are up to 300 μm long and elongate, with aspect ratios up to 6:1. Most exhibit concentric zoning and some crystals contain metamict zones (Fig. 5*d*). Thirteen analyses of 13 zircons are concordant to slightly discordant (Fig. 6d). Most analyses indicate high uranium and thorium contents (Table 2). One analysis that yields high within-run variation of isotope ratios is not considered further. A regression from initial Pb ($^{207}Pb/^{206}Pb = 0.8974$ at 861 Ma) through uncorrected data yields a concordia intercept at 861 ± 10 Ma (MSWD = 1.7) for eight analyses. Four slightly younger results (822–813 Ma) are interpreted to reflect minor loss of radiogenic Pb. The date of 861 ± 10 Ma is interpreted as the crystallization age of this sample.

U–Th–Pb analyses of zircons from granite-porphyry sample A-10-4-2 were performed using the SHRIMP II ion microprobe in the Center for Isotopic Investigations at VSEGEI (St. Petersburg), following the procedure described in Larionov et al. (2004) and Vernikovsky et al. (2007).

Zircons from porphyritic granite sample A-10-4-2 are subhedral to euhedral, colorless and semicolorless. There are also fragments of colorless and yellowish elongate crystals. The crystals are from 100 to 300 μm long, with aspect ratios from 1.5:1 to 3.0:1. Cathodoluminescence structure, idiomorphism, and magmatic zoning indicate their magmatic origin. Only euhedral zircons without defined cores, fractures and inclusions were selected for analysis using optical, Cl and BSE images. Eleven analyses of zircons were conducted (Table 3).

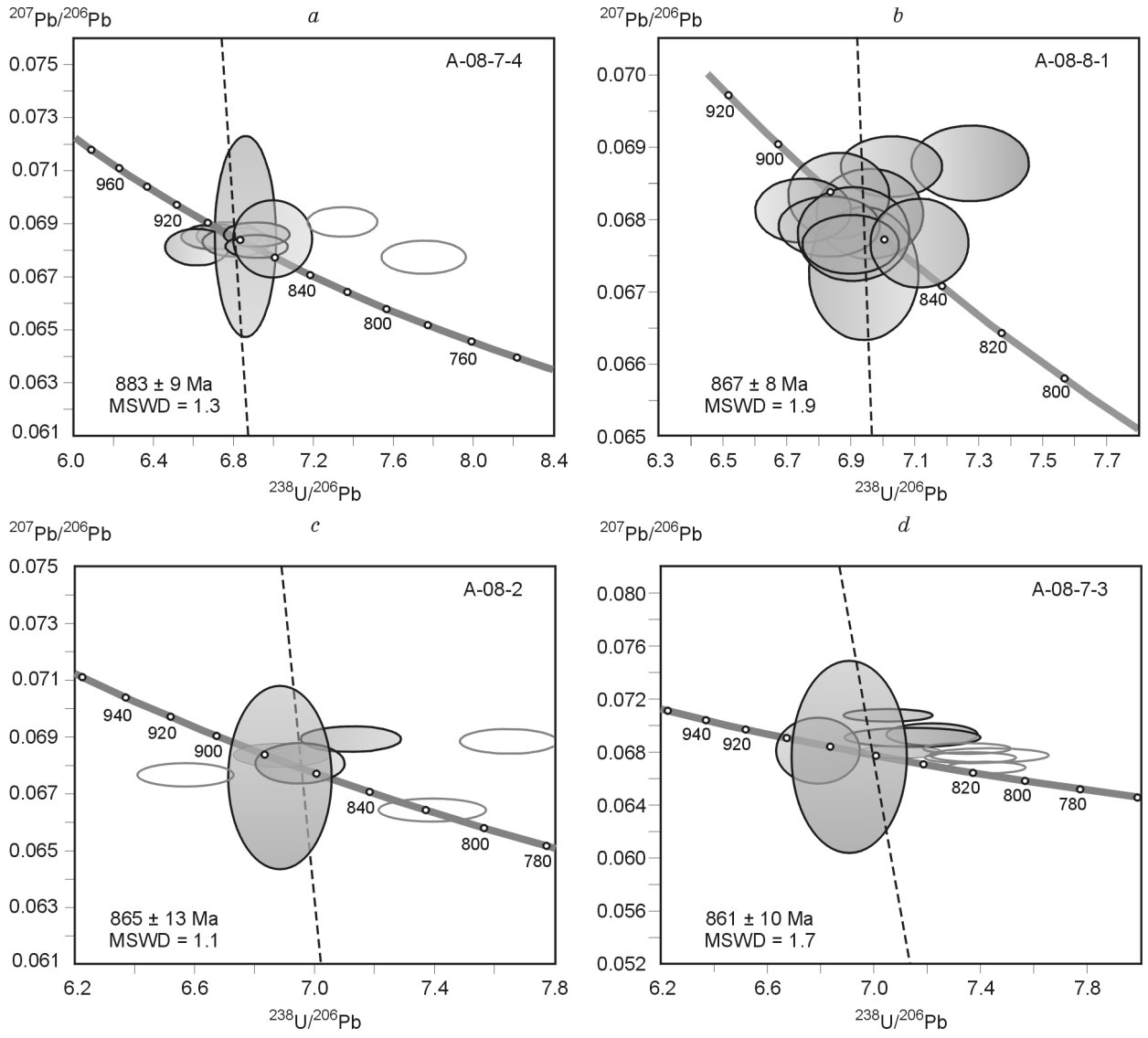


Fig. 6. Concordia diagrams for zircons from samples A-08-7-4 (*a*), A-08-8-1 (*b*), A-08-2 (*c*), and A-08-7-3 (*d*) from the Yeruda pluton in the region between the Chirimba and Yeruda Rivers.

Eight analyses form a concordant group (Fig. 7), corresponding to a date of 880.0 ± 6.5 Ma (MSWD = 0.063), which is interpreted as the age of crystallization. Two results with high U (2576 and 2034 ppm), and a third with high common Pb (1.41% common ^{206}Pb), yield younger dates of 846–786 Ma, interpreted to reflect loss of radiogenic Pb.

Paleomagnetism. During paleomagnetic investigations special attention was given to the composition and structure of magnetic minerals and their distribution in the rocks. During fieldwork the most strongly magnetized samples were selected using measurements of magnetic susceptibility with a KT-5 field meter (AGICO, Czech Republic). Core samples were taken using a portable petrol-powered drill, and oriented by both magnetic and solar compass. Both partially altered and unaltered variations of the rocks were sampled. Petromagnetic and paleomagnetic laboratory experiments were performed in the Laboratory of Geodynamics and Paleomagnetism of IPGG SB RAS using standard procedures (Tauxe, 2010). All samples underwent alternating magnetic field (AF) demagnetization within a 2G Enterprises R-755 cryogenic magnetometer housed in a magnetically shielded room. Determination of stable magnetization directions was performed using orthogonal diagrams (Zijderveld, 1967) and principal component analysis (Enkin, 1994; Kirschvink, 1980).

Granites are challenging to study with paleomagnetic methods. On one hand the lengthy process of acquiring a thermoremanent magnetization (TRM) (up to 40 ka in one sample) during cooling of the granite plutons provides an averaging of the secular variations of the geomagnetic field (Wakabayashi, 2006). On the other hand the structural attitude (particularly the paleohorizontal at the time of magnetization) of granitic bodies is of course not always known, and failing to account for subsequent deformations can lead to inaccurate interpretation of paleomagnetic data. Additional complications are related to granite magnetic mineralogy, mostly because of the lack of the main TRM carrier, magnetite. Consequently

Table 2. U–Th–Pb analytical data for zircons from igneous rocks of the Yeruda pluton

Grain zone	Content, ppm		Isotope ratio						Age (Ma)		*D*, %
	^{238}U	^{232}Th	$^{232}Th/^{238}U$	f_{204}, %	$^{238}U/^{206}Pb$, ±1σ	$^{207}Pb/^{206}Pb$, ±1σ	$^{238}U/^{206}Pb^*$, ±1σ	$^{207}Pb/^{206}Pb^*$, ±1σ	$^{238}U/^{206}Pb^*$, ±1σ	$^{207}Pb^*/^{206}Pb^*$, ±1σ	
Sample A-08-7-4, granodiorite											
1.1	1803	522	0.30	0.706	17.84 ± 0.259	0.05596 ± 0.00871	17.96 ± 0.262	0.05032 ± 0.00880	349 ± 5	210 ± 405	–66.5
2.1	628	16	0.03	–0.046	6.856 ± 0.102	0.06848 ± 0.00252	6.853 ± 0.102	0.06887 ± 0.00253	878 ± 12	895 ± 76	1.9
3.1	970	664	0.71	0.175	7.343 ± 0.117	0.06903 ± 0.00037	7.355 ± 0.117	0.06759 ± 0.00046	822 ± 12	856 ± 14	4.0
4.1	953	456	0.49	0.034	7.751 ± 0.140	0.06769 ± 0.00043	7.754 ± 0.140	0.06741 ± 0.00045	782 ± 14	850 ± 14	8.1
5.1	1919	272	0.15	0.083	6.771 ± 0.106	0.06851 ± 0.00035	6.777 ± 0.106	0.06782 ± 0.00037	887 ± 13	863 ± 11	–2.8
6.1	563	193	0.35	–0.021	6.614 ± 0.106	0.06808 ± 0.00046	6.612 ± 0.106	0.06825 ± 0.00048	908 ± 14	876 ± 14	–3.6
7.1	1117	963	0.89	0.050	6.810 ± 0.111	0.06827 ± 0.00037	6.813 ± 0.111	0.06786 ± 0.00039	883 ± 14	864 ± 12	–2.1
8.1	131	63	0.50	0.216	6.995 ± 0.131	0.06839 ± 0.00097	7.010 ± 0.132	0.06660 ± 0.00132	860 ± 15	825 ± 41	–4.2
9.1	2277	1215	0.55	0.006	6.913 ± 0.103	0.06810 ± 0.00028	6.913 ± 0.103	0.06806 ± 0.00029	871 ± 12	870 ± 9	–0.1
10.1	1478	845	0.59	0.000	6.695 ± 0.106	0.06847 ± 0.00032	6.695 ± 0.106	0.06847 ± 0.00032	897 ± 14	883 ± 10	–1.6
11.1	1828	1248	0.70	0.007	6.912 ± 0.109	0.06855 ± 0.00031	6.913 ± 0.109	0.06850 ± 0.00031	871 ± 13	884 ± 9	1.4
Sample A-08-8-1, quartz diorite											
1.1	1429	676	0.49	0.007	6.749 ± 0.099	0.06811 ± 0.00030	6.749 ± 0.099	0.06805 ± 0.00030	891 ± 12	870 ± 9	–2.4
2.1	2039	1383	0.70	–0.010	6.897 ± 0.100	0.06764 ± 0.00027	6.897 ± 0.100	0.06773 ± 0.00027	873 ± 12	860 ± 8	–1.5
3.1	1073	176	0.17	0.000	7.272 ± 0.122	0.06877 ± 0.00035	7.272 ± 0.122	0.06877 ± 0.00035	831 ± 13	892 ± 10	6.9
4.1	500	228	0.47	–0.063	6.907 ± 0.116	0.06779 ± 0.00043	6.902 ± 0.115	0.06831 ± 0.00048	872 ± 14	878 ± 14	0.7
5.1	635	157	0.26	0.050	6.957 ± 0.111	0.06807 ± 0.00042	6.961 ± 0.111	0.06765 ± 0.00045	865 ± 13	858 ± 14	–0.9
6.1	1932	772	0.41	0.025	6.830 ± 0.105	0.06789 ± 0.00028	6.831 ± 0.105	0.06768 ± 0.00029	881 ± 13	859 ± 9	–2.5
7.1	2170	1285	0.61	0.114	7.025 ± 0.105	0.06872 ± 0.00028	7.033 ± 0.105	0.06778 ± 0.00032	857 ± 12	862 ± 10	0.6
8.1	283	76	0.28	0.101	6.938 ± 0.113	0.06724 ± 0.00061	6.945 ± 0.113	0.06640 ± 0.00072	867 ± 13	819 ± 23	–5.9
9.1	775	133	0.18	0.029	6.859 ± 0.105	0.06832 ± 0.00039	6.861 ± 0.105	0.06809 ± 0.00041	877 ± 13	871 ± 13	–0.7
10.1	799	552	0.71	–0.017	7.113 ± 0.102	0.06766 ± 0.00041	7.111 ± 0.102	0.06780 ± 0.00042	848 ± 12	862 ± 13	1.6
Sample A-08-2, plagiogranite											
1.1	489	337	0.71	0.035	3.120 ± 0.045	0.10898 ± 0.00051	3.121 ± 0.045	0.10868 ± 0.00053	1792 ± 23	1777 ± 9	–0.8
2.1	1528	1183	0.80	0.130	7.123 ± 0.411	0.07277 ± 0.01473	7.132 ± 0.412	0.07168 ± 0.01475	846 ± 48	977 ± 419	13.4
3.1	3109	2289	0.76	0.035	7.393 ± 0.119	0.06637 ± 0.00027	7.395 ± 0.119	0.06608 ± 0.00028	818 ± 12	809 ± 9	–1.1
4.1	642	53	0.09	0.012	6.949 ± 0.098	0.06802 ± 0.00047	6.950 ± 0.098	0.06792 ± 0.00048	867 ± 12	866 ± 15	–0.1
5.1	1654	31	0.02	0.354	7.279 ± 3.246	0.07215 ± 0.00035	7.305 ± 3.258	0.06920 ± 0.00051	827 ± 596	905 ± 15	8.6
6.1	2448	1315	0.55	0.303	7.648 ± 0.107	0.06880 ± 0.00029	7.671 ± 0.108	0.06629 ± 0.00040	790 ± 11	816 ± 13	3.1
7.1	2219	1827	0.85	0.047	6.567 ± 0.106	0.06761 ± 0.00028	6.570 ± 0.106	0.06722 ± 0.00030	913 ± 14	845 ± 9	–8.1
8.1	1446	617	0.44	0.166	6.202 ± 0.100	0.07696 ± 0.00036	6.212 ± 0.100	0.07557 ± 0.00043	962 ± 15	1084 ± 11	11.2
9.1	237	128	0.56	0.436	3.539 ± 0.055	0.15422 ± 0.00905	3.555 ± 0.056	0.15034 ± 0.00912	1598 ± 23	2350 ± 104	32.0
10.1	1188	927	0.81	0.419	6.000 ± 0.090	0.08726 ± 0.00039	6.025 ± 0.090	0.08370 ± 0.00054	990 ± 14	1286 ± 13	23.0
11.1	1687	868	0.53	0.030	6.889 ± 0.107	0.06832 ± 0.00029	6.891 ± 0.107	0.06807 ± 0.00030	874 ± 13	871 ± 9	–0.3
12.1	300	247	0.85	–0.013	4.430 ± 0.072	0.08792 ± 0.00060	4.430 ± 0.072	0.08804 ± 0.00062	1312 ± 20	1383 ± 13	5.1
13.1	1910	1202	0.65	0.078	7.120 ± 0.111	0.06889 ± 0.00030	7.125 ± 0.111	0.06824 ± 0.00033	847 ± 13	876 ± 10	3.3
11.2	1951	1262	0.67	0.051	6.883 ± 0.114	0.06751 ± 0.00216	6.886 ± 0.114	0.06709 ± 0.00216	874 ± 14	840 ± 67	–4.0
Sample A-08-7-3, leucogranite											
1.1	3317	1775	0.55	0.019	7.947 ± 1.196	0.06597 ± 0.00343	7.949 ± 1.197	0.06581 ± 0.00344	764 ± 126	800 ± 109	4.5
2.1	1102	151	0.14	0.027	6.905 ± 0.143	0.06756 ± 0.00485	6.907 ± 0.143	0.06734 ± 0.00485	872 ± 17	848 ± 150	–2.8
3.1	1154	368	0.33	0.021	6.787 ± 0.104	0.06804 ± 0.00166	6.788 ± 0.104	0.06786 ± 0.00167	886 ± 13	864 ± 51	–2.5
4.1	2315	605	0.27	0.054	7.348 ± 0.110	0.06819 ± 0.00027	7.352 ± 0.110	0.06774 ± 0.00029	822 ± 12	861 ± 9	4.5
5.1	2176	750	0.36	0.048	7.431 ± 0.148	0.06767 ± 0.00038	7.435 ± 0.148	0.06728 ± 0.00040	813 ± 16	846 ± 12	3.9
6.1	1407	249	0.18	0.054	7.143 ± 0.169	0.06903 ± 0.00048	7.146 ± 0.169	0.06859 ± 0.00051	844 ± 19	886 ± 15	4.7
7.1	1573	377	0.25	0.254	7.050 ± 0.111	0.07074 ± 0.00033	7.068 ± 0.111	0.06862 ± 0.00043	853 ± 13	887 ± 13	3.9
8.1	2411	471	0.20	0.026	6.928 ± 0.104	0.06774 ± 0.00028	6.930 ± 0.104	0.06752 ± 0.00029	869 ± 12	854 ± 9	–1.8
9.1	1495	876	0.61	–0.011	6.903 ± 0.108	0.06783 ± 0.00032	6.902 ± 0.108	0.06792 ± 0.00032	872 ± 13	866 ± 10	–0.7
10.1	2329	804	0.36	0.100	7.361 ± 0.113	0.06750 ± 0.00028	7.368 ± 0.113	0.06667 ± 0.00031	820 ± 12	828 ± 10	0.9
11.1	1968	324	0.17	0.062	7.417 ± 0.100	0.06675 ± 0.00029	7.422 ± 0.100	0.06624 ± 0.00032	815 ± 10	814 ± 10	–0.1
12.1	1297	630	0.50	0.038	7.100 ± 0.105	0.06793 ± 0.00033	7.103 ± 0.105	0.06761 ± 0.00035	849 ± 12	857 ± 11	0.9
13.1	361	166	0.48	0.144	7.216 ± 0.114	0.06928 ± 0.00056	7.226 ± 0.114	0.06808 ± 0.00069	836 ± 13	871 ± 21	4.1

Note. Grain zone: the first number indicates the analyzed grain, the second—analyses within a grain. f_{204}, Common ^{206}Pb content in total measured ^{206}Pb, corrected for measured ^{204}Pb. Pb*, Radiogenic lead Pb.

Table 3. Results of U/Th/Pb isotopic investigations of zircons from a Yeruda pluton granite-porphyry

Grain zone	Content, ppm			Isotope ratio						Err. corr.	Age, Ma		*D*, %
	$^{206}Pb^*$	U	Th	$^{232}Th/^{238}U$	% $^{206}Pb_c$	(1) $^{238}U/^{206}Pb^*$, ±%	(1) $^{207}Pb^*/^{206}Pb^*$, ±%	(1) $^{207}Pb^*/^{235}U$, ±%	(1) $^{206}Pb^*/^{238}U$, ±%		(1) $^{206}Pb/^{238}U$, ±%	(1) $^{207}Pb/^{206}Pb$, ±%	
Sample A-10-4-2													
1.1	48.7	396	161	0.42	0.04	7.0 ± 1.1	0.0688 ± 1.2	1.36 ± 1.6	0.143 ± 1.1	0.7	863 ± 9	893 ± 24	+4
2.1	33	296	109	0,38	1.41	7.7 ± 1.2	0.0660 ± 3.5	1.18 ± 3.7	0.130 ± 1.2	0.3	786 ± 9	805 ± 73	+3
2.2	310	2576	302	0,12	0.02	7.1 ± 1.0	0.0676 ± 0.5	1.31 ± 1.1	0.140 ± 1.0	0.9	846 ± 8	857 ± 10	+1
3.1	47.7	382	119	0.32	–	6.9 ± 1.2	0.0687 ± 1.2	1.38 ± 1.7	0.146 ± 1.2	0.7	876 ± 10	890 ± 24	+2
4.1	25.3	956	401	0.43	0.02	6.8 ± 1.3	0.0664 ± 3.5	1.35 ± 3.7	0.148 ± 1.3	0.3	887 ± 11	820 ± 73	–9
5.1	26.6	375	15	0.04	0.22	6.8 ± 1.2	0.0681 ± 1.6	1.38 ± 2.0	0.147 ± 1.2	0,6	882 ± 10	870 ± 34	–1
6.1	18.1	1059	102	0.10	0.02	6.9 ± 1.1	0.0693 ± 0.7	1.39 ± 1.3	0.145 ± 1.1	0,8	875 ± 9	908 ± 14	+4
7.1	28.5	775	46	0.06	0.02	6.8 ± 1.1	0.0673 ± 1.0	1.37 ± 1.5	0.147 ± 1.1	0.7	887 ± 9	847 ± 21	–5
7.2	12.8	2034	255	0.13	0.02	7.2 ± 1.1	0.0668 ± 0.5	1.27 ± 1.2	0.138 ± 1.1	0.9	836 ± 8	830 ± 11	–1
8.1	39.3	271	83	0.32	0.07	6.7 ± 2.2	0.0655 ± 4.2	1.34 ± 4.8	0.149 ± 2.2	0.5	895 ± 19	791 ± 88	–14
8.1re	12.1	911	396	0.45	0.04	6.8 ± 1.1	0.0681 ± 0.8	1.39 ± 1.4	0.148 ± 1.1	0.8	888 ± 9	871 ± 16	–2

Note. Uncertainties are given at the 1σ level; 1σ calibration errors of the TEMORA standard for sample A-10-4-2 are 0.28%; $^{206}Pb_c$ and $^{206}Pb^*$ are common and radiogenic lead, respectively; (1), common lead, corrected for measured ^{204}Pb. *D*, %, Discordance.

the suitability of granites is determined by the size of magnetite grains and the degree of secondary alterations. For Precambrian granites, this situation is worsened by their long and typically complex geological history and poor preservation. However, detailed studies of magnetic mineralogy together with reliable structural constraints have led to successful paleomagnetic investigations of Precambrian granites (Ballard et al., 1989; Didenko et al., 2005, 2006; Gregory et al., 2009; Meert, 2002). In these cases the studied granite plutons did not undergo significant deformation (dips of <10°), and the magnetization carrier is SD or pseudo-single-domain (PSD) magnetite or low-titanium titanomagnetite.

Despite low magnetization intensities, these samples can be accurately measured using modern instruments. Natural remanent magnetization (NRM) intensities are 0.07–1.2 mA/m. Magnetic susceptibility (χ) varies from 50 to 300×10^{-6} SI units in rocks of the main Yeruda pluton and from 2 to 200×10^{-6} SI units in the dikes. Significant variations of χ in the dikes can be explained by heterogeneity of the initial distribution of magnetic minerals in the rock during emplacement (Pechersky and Didenko, 1995). The Q-value (NRM/χ × Ho) was measured at 0.01–0.3, except for individual samples with low χ.

Thermomagnetic analysis results (Lowrie, 1990) indicate the dominance of magnetically soft iron-bearing minerals (Fig. 8*a*). In almost all studied samples, unblocking temperature (T_{ub}) intervals of 150–550 °C correspond to partially oxidized minerals of the magnetite-ulvospinel series ($FeFe_2O_4$–Fe_2TiO_4). However, the presence of high-iron titanomagnetite to magnetite is well defined in χ(*T*) diagrams; diagnostic characteristics include dome-shaped curves, abrupt decrease of susceptibility at heating to 500–550 °C, and growth during cooling (Fig. 8*b*). To determine the domain state of titanomagnetite we plotted a Day-Dunlop diagram (Fig. 9; Day et al., 1977; Dunlop, 2002), which indicates that most of the studied samples are single-domain (SD) or pseudo-domain (PSD) (grain sizes 0.1–20 μm) (Dunlop and Özdemir, 1997). Typically such a domain state is typical for titanomagnetite of magmatic origin, whereas the state of secondary magnetite and hematite grains is usually SD (Nédélec et al., 2015). The discrepancy between the points and the theoretical curve is due to the occurrence of magnetic fragments of another chemical composition in addition to titanomagnetite. According to the Lowrie thermomagnetic analysis (Fig. 8), these are magnetically hard phases, including hematite (T_{ub} ~ 675 °C) and probably maghemite (T_{ub} ~ 150 °C) that are probably secondary alteration minerals. Samples with high M_{rs}/M_s ratios (0.45–0.55) can contain secondary superparamagnetic grains.

Our petromagnetic investigations indicate that the primary paleomagnetic signal in the granites is related to relatively large PSD titanomagnetite grains, close to the PAS—multidomain boundary, which have a thermoremanent or partial thermoremanent magnetization (PTRM) but are characterized by low coercivity. Consequently the age of magnetization of the NRM component at a first approximation will correspond to the U–Pb zircons age estimates (883–861 Ma).

Alternating magnetic (AF) field demagnetization resulted in the successful isolation of a low-coercivity component corresponding to titanomagnetite in most of the studied samples. Typically this component is destroyed by a ~30 mT field, and the median destructive field (MDF) is 5 to 20 mT (Fig. 10). Complete destruction of the NRM is accomplished by a relatively low AF of 50–70 mT or in some cases up to 100 mT if the sample contains a magnetically hard secondary mineral (possibly maghemite), that exhibits random magnetic directions in contrast to the low-coercivity component.

The Fisher precision parameter of the low-coercivity component vectors is not high in individual outcrops—usually less

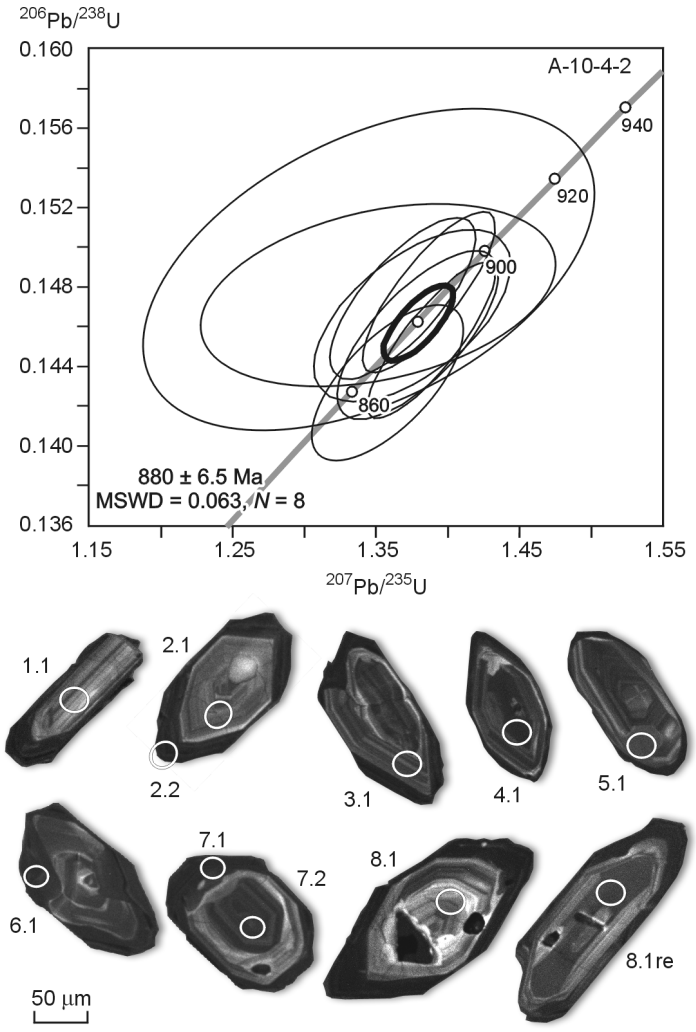


Fig. 7. Concordia diagrams and CL images for zircons from granite-porphyry sample A-10-4-2 of the Yeruda pluton the region between the Chirimba and Yeruda Rivers.

than 10; and α_{95} values are up to 25°. Parameters for partially altered variations of granite samples are somewhat higher (Table 4). Considering the weakness of the paleomagnetic signal, this is a reflection of the low magnetization intensities. By increasing the statistics and averaging all identified directions for single vectors as well as for vectors previously grouped in outcrops we are able to evaluate a mean paleomagnetic direction (D, I) of 315.3°, 67.7°, $\alpha_{95} = 5.6°$, and a paleopole at 63.5° N, 3.6° E, $A_{95} = 8.6°$, for the Yeruda pluton with a low error parameter (Table 4, Fig. 11).

The U–Pb zircon crystallization ages for five samples range from 883 to 861 Ma, with a mean age of about 873 Ma. The calculated paleopole is assumed, due to slow cooling, to have adequately averaged paleosecular variation (PSV) and therefore is considered to reflect accurately the magnetic pole at the time of pluton emplacement. The paleopole is significantly different from the corresponding Neoproterozoic interval of the apparent polar wander path (APWP) of Siberia (Fig. 11). Considering the uncertainties and the lack of information concerning the polarity of the geomagnetic field at the time of pluton emplacement, the discrepancy in paleolatitudes between the position of the Yeruda pluton and the Siberian craton would have been at least 8.6°. The northernmost position for the pluton in stratigraphic coordinates, considering the error parameter and assuming a reversed geomagnetic field polarity would have been 51.5°–9.7° = 41.8° S, and the southernmost expected position for the craton, accounting for the error parameters would have been 24.7° + 8.5° = 33.2° S. This indicates that at 880–860 Ma, during the tectonic event that formed the Teya complex granites, the Central Angara terrane was located at a distance of no less than one thousand kilometers from Siberia. These paleomagnetic results render indefensible the conclusion by some researchers (Likhanov et al., 2014; Nozhkin et al., 2015) of "Grenville-age orogenesis" within the Yenisei Ridge.

The estimates presented in this study are correct if the determined paleomagnetic directions were acquired during a time of reverse polarity of the geomagnetic field. In this case, the Central Angara terrane must have been located in moderate southern latitudes (–41.8° to –61.2°). According to modern tectonic reconstructions for 880 Ma, these latitudes were occupied by the pre-Timanian margin of Baltica, which was

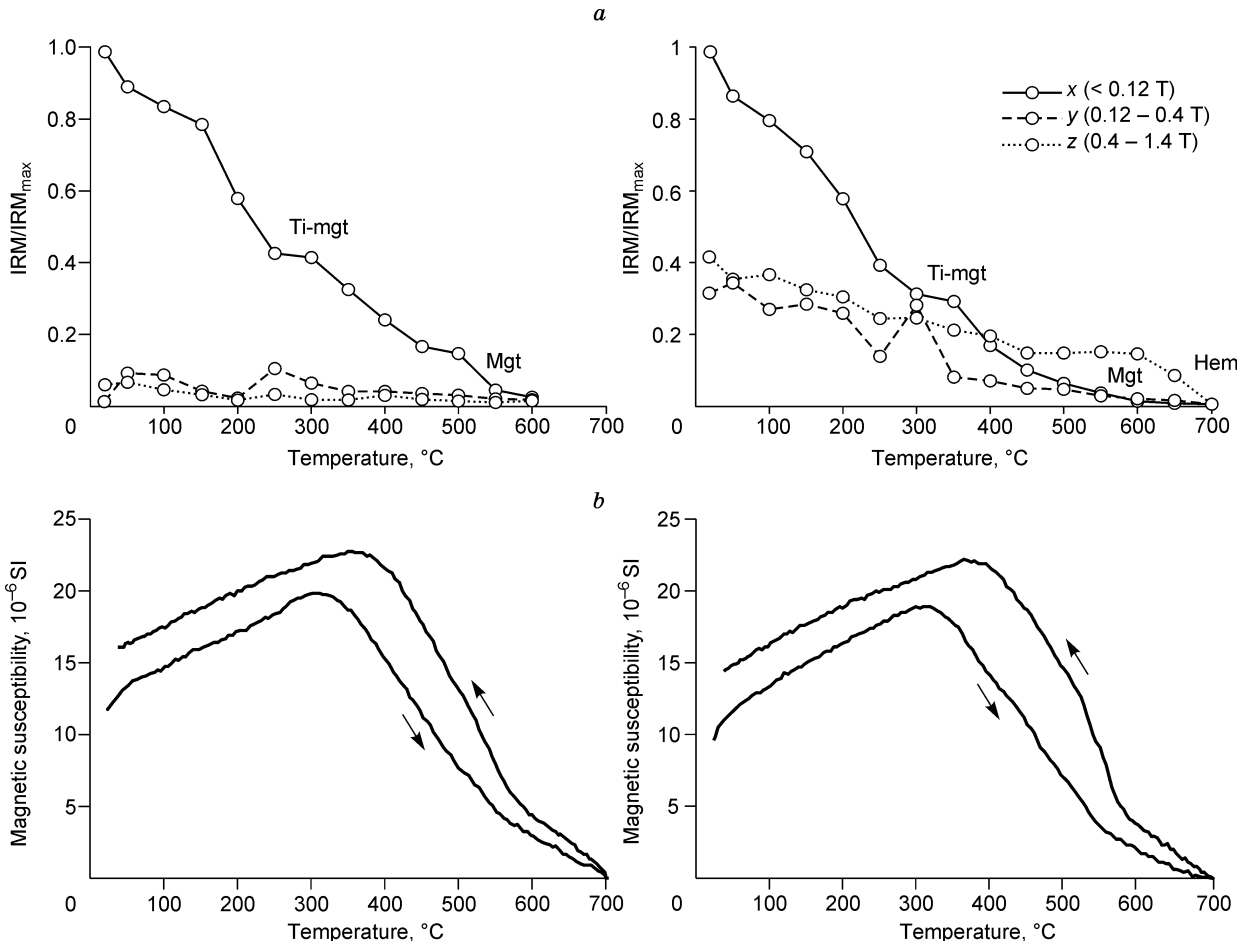


Fig. 8. Thermomagnetic analysis results for Yeruda pluton granites. (*a*), examples of isothermal remanent magnetization (IRM) vs. temperature (Lowrie, 1990); (*b*), examples of magnetic susceptibility vs. temperature.

part of Rodinia (Metelkin et al., 2007, 2015; Vernikovsky et al., 2009), and where subduction and accretion were probably the dominant processes (Gee et al., 2006). Such a geodynamic setting is in agreement with the geochemical nature of the Teya granitoids. Therefore, we propose that the formation of the Yeruda pluton and its analogues was caused by collision of the Central Angara terrane with an island arc at the margin of Baltica. However, considering the growth of the ocean between Baltica and Siberia at this time, the kinematics of the subsequent drift of the Central Angara terrane towards its current "Siberian" position may have been very complex.

During the 100–120 Ma interval that separates the Yeruda magmatism and the collision between the Central Angara terrane and the East Angara margin of Siberia, which as we believe is marked by the emplacement of the Ayakhta granites, the Central Angara terrane should have drifted northwards for $42.2° \pm 8.5°$ (about 4500 km) while "outrunning" the Siberian paleocontinent. The calculated mean drift velocity of the Central Angara terrane would have been 4 to 6 cm/year. For the sake of comparison, the velocity of the Siberian plate inferred from paleomagnetic data for the same time interval was no more than 3 cm/year and reached comparable values only in the 800–780 Ma interval (Metelkin et al., 2007). At the same time, this northwards drift of the Central Angara

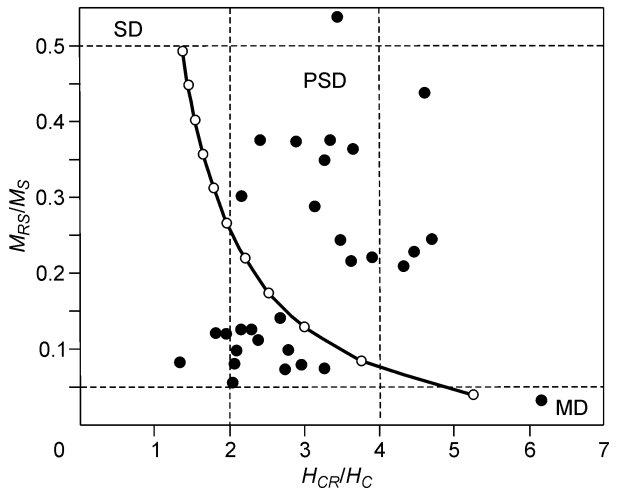


Fig. 9. Day diagram (Day et al., 1977) demonstrating the connection between the hysteresis parameter ratios. M_{RS}, saturation remanent magnetization / M_S, saturation magnetization; H_{CR}, remanent coercive force / H_C, coercive force for multidomain (MD), pseudosingle-domain (PSD) and single-domain (SD) magnetic grains. The curve with white circles is a theoretical line plotted for a mix of SD and MD titanomagnetite grains (Dunlop, 2002).

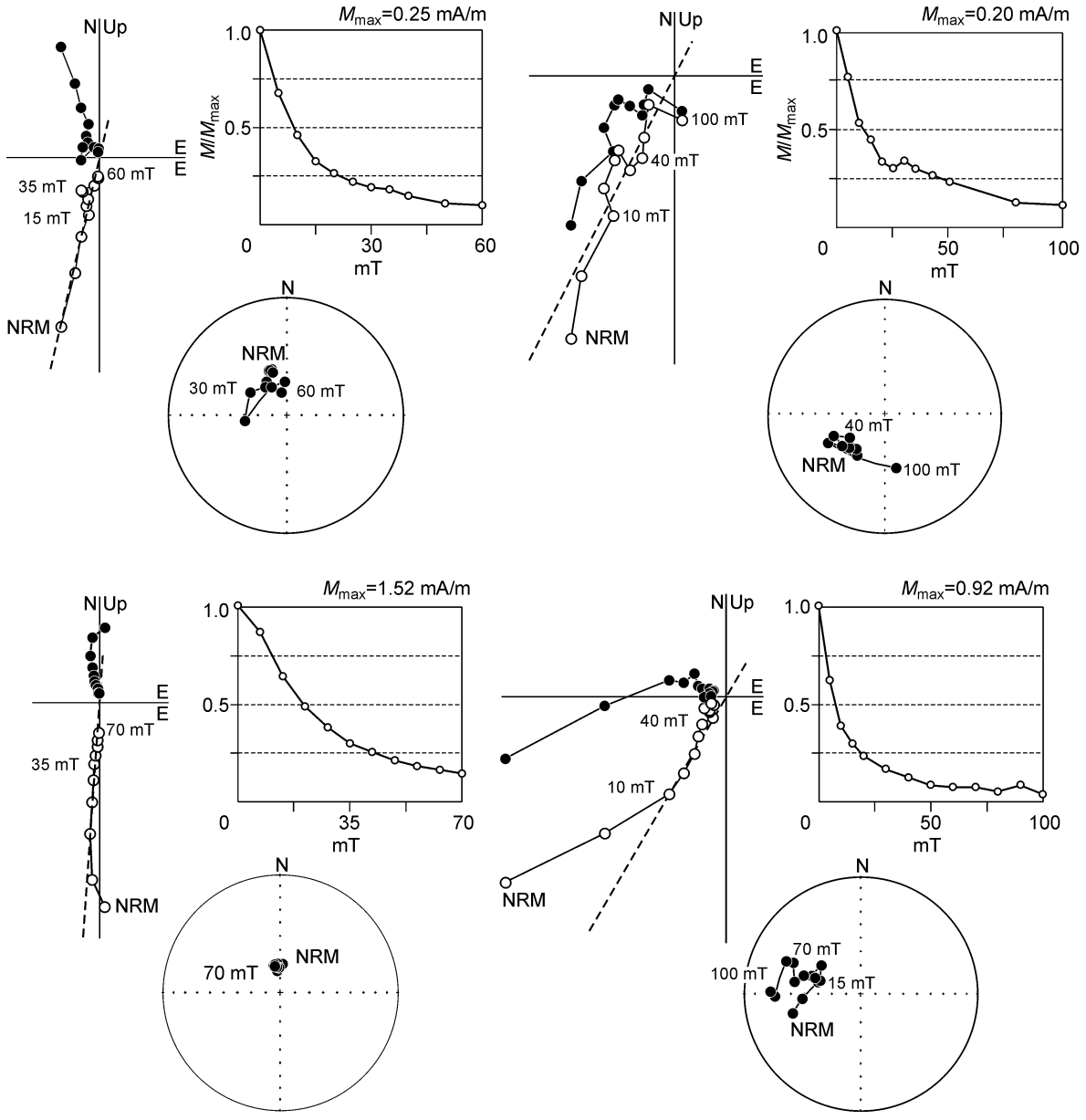


Fig. 10. Typical orthogonal plots, stereoplots (equal-area projections) and NRM/H graphs for results of alternating field demagnetization. Closed (open) circles on orthogonal plots are magnetization vector endpoints in the horizontal (vertical) plane. Dotted line shows the direction of the magnetization regular component.

terrane must have been accompanied by a significant rotation around a vertical axis. The rotation angle would have been 161.7° ± 12.6°; therefore, the mean rotation rate would have been 2°/Ma, while the rotation of the Siberian plate was no more than 0.5°/Ma.

If we suppose that the Yeruda pluton magnetization corresponds to a normal polarity of the geomagnetic field and the Central Angara terrane was located in the northern hemisphere, then the distance to its present-day position on the East Angara margin of Siberia would have been more than 7500 km, with an angular distance between the mean Yeruda pole and the 880 Ma pole for Siberia is 76.3° ± 8.7° (Fig. 11). For the 100–120 Ma prior to its collision with Siberia, the Central Angara terrane should have travelled a distance of 61.1° ± 8.5° (about 6000 km), with a calculated drift speed of 6–7 cm/year. At the same time, the rotation of the Central Angara terrane would have been only 18.3° ± 12.6°. Considering that the Siberian plate would have been moving towards the Central Angara terrane, such a tectonic scenario appears realistic despite the large distance.

In addition, paleomagnetic data we obtained recently for metarhyolites of the Central Taimyr volcanic-plutonic association (Proskurnin et al., 2014) agree with this supposition. These volcanic rocks in association with the Snezhnin (Zhdanov) complex granitoids formed an extensive belt 870–820 Ma, as a result of collision of an island arc located on the periphery of the Siberian paleocontinent with large continental terranes (Faddey and Mamont-Shrenk) that were part of the Central Taimyr accretionary belt (Proskurnin et al., 2014). The geochemical composition of these granites of the

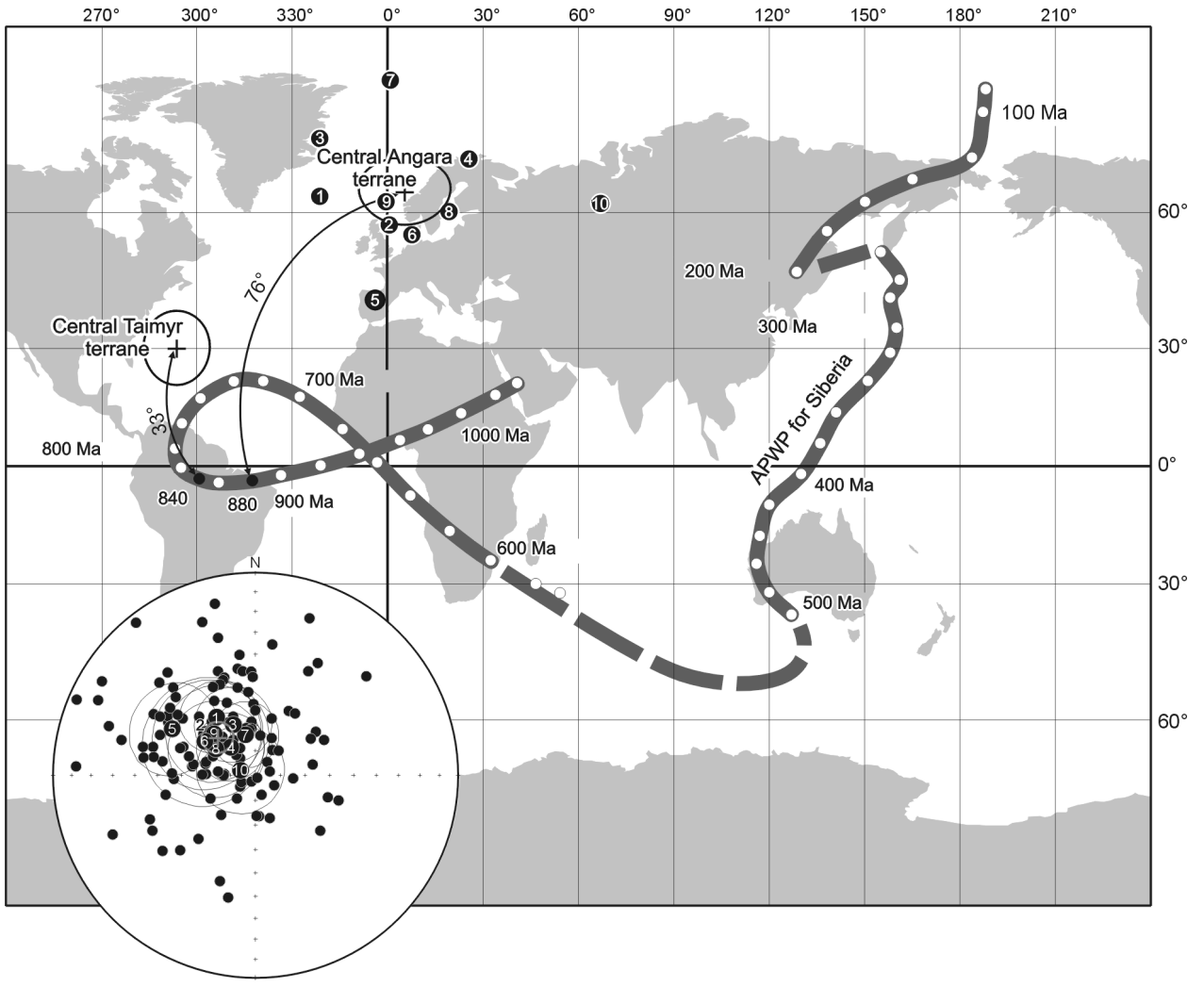


Fig. 11. Distribution of paleomagnetic directions (stereogram) and paleomagnetic poles (Table 4) for the Yeruda granites in relation to the APWP for Siberia (from Metelkin et al., 2012). Numbers in circles correspond to numbers of sampling sites from Fig. 2 and Table 1.

Central Taimyr volcanic-plutonic association is similar to that of the Teya complex. Although the paleomagnetic directions obtained for the metarhyolites are dated at ~850 Ma and cannot be directly compared with those in the Yeruda granites, the paleomagnetic pole is very close to Siberia (Fig. 11). According to these data (Table 4) the accretionary events that caused the formation of the rhyolite-granite association also took place at a considerable distance in the north, away from the continental margin of Siberia. Thus, it is not excluded that at 880 Ma the Central Angara terrane could have been in the same tectonic unit as the continental blocks (Faddey, Mamont-Shrenk terranes) that are now part of the Central Taimyr accretionary belt.

Discussion of results and conclusions

We have already published several demonstrations that a Grenville-age orogenic belt could not have existed along the western margin of the Siberian craton (Li et al., 2008; Pisarevsky et al., 2008; Vernikovsky and Vernikovskaya, 2006; Vernikovsky et al., 2003, 2007, 2009). Nonetheless, some researchers insist on maintaining the opposite viewpoint, based mainly on findings of Mesoproterozoic detrital zircons in gneisses and granite-gneisses of the Yenisei Ridge (Likhanov and Reverdatto, 2014; Likhanov et al., 2014; Nozhkin et al., 2008b, 2015) even though they cannot actually be considered evidence of a collisional event. At the same time, these authors do not take geological data into account. For example, analyses of the Meso-Neoproterozoic sedimentary complexes of the western framing of the Siberian paleocontinent (in geographic coordinates) demonstrated that they are typical of passive continental margins (Khabarov, 2011; Petrov and Semikhatov, 1997; Pisarevsky and Natapov, 2003; Pisarevsky et al., 2008; Sergeev et al., 1997; Zonenshain et al., 1990). However, a passive continental margin setting is indicative of a transitional zone from continent to ocean and not of a collisional belt. Additionally, analyses of the structure of the sedimentary deposits are in complete agreement with the abundant data on the age of igneous complexes of the Yenisei Ridge. According to these data, between the formation of the hypersthene granites (kuzeevites) of the Angara–Kan

Table 4. Paleomagnetic directions, paleopoles coordinates and paleolatitudes for the studied Yeruda pluton granites in comparison to data for Siberia and the Central Taimyr accretionary belt

Sampling site	GPS coordinates	Rock type	n	Paleomagnetic directions				Paleopole		A_{95}	PL
				D	I	k	α_{95}	Latitude	Longitude		
11	N 59° 51′ 25.6″ E 93° 20′ 27.1″	Granite	11	326.2	61.3	10.6	14.7	63.1	338.7	19.8	42.4
12	N 59° 40′ 58.8″ E 93° 29′ 02.6″	Granodiorite	17	310.0	63.9	3.6	22.0	57.5	0.5	31.2	45.6
13		Granite-porphyry	16	336.2	67.3	6.7	15.4	73.4	338.7	23.3	50.1
14		Granite	14	318.7	74.6	9.3	13.7	69.9	25.6	23.7	61.1
15	N 59° 40′ 44.7″ E 93° 28′ 13.5″	Leucogranite	14	299.3	50.8	5.8	18.1	41.4	356.1	20.1	31.5
5	N 59° 41′ 33.8″ E 93° 26′ 34.4″	Granite	8	304.0	65.1	5.6	25.8	55.6	7.6	37.5	47.1
5		Granite	8	345.7	73.0	18.0	13.4	82.6	0.8	22.6	58.6
7	N 59° 41′ 33.5″ E 93° 26′ 30.4″	Granodiorite	11	303.8	70.6	6.5	19.4	60.2	19.3	31.3	54.8
7		Granodiorite	10	315.4	66.0	7.1	19.5	62.1	359.5	28.8	48.3
6		Leucogranite	9	290.5	83.4	9.4	17.7	61.8	66.9	34.4	77.0
Average for samples			118	315.3	67.7	6.5	5.6	63.5	3.6	8.6	50.6
Average for sites			10	315.0	68.3	59.7	6.3	63.9	5.4	9.7	51.5
Expected, Siberia*			–	310.3	–42.6	–	8.8	–3.7	317.5	8.5	–24.7
Metarhyolites of the Central Taimyr volcanic-plutonic association**			22	351.0	29.9	12.8	9.0	29.9	293.6	7.4	16.0

Note. Sequence numbers correspond to sampling sites on Fig. 2 and Table 1; n, number of vectors accepted for calculation; D, declination in degrees; I, inclination in degrees; k, precision parameter; α_{95}, 95% error parameter; A_{95}, 95% error parameter for paleopole and paleolatitudes; PL, paleolatitude for the block in the reconstruction in degrees north latitude.

* Expected direction is calculated for coordinates of the Yeruda pluton sampling site (59.7° N, 93.5° E) according to the paleomagnetic pole for 880 Ma in the Siberia APWP (Metelkin et al., 2012).

** Data for 850 Ma from (Proskurnin et al., 2014), further comments in the text.

uplift at 1734 Ma (Bibikova et al., 2001) and the oldest granites of the Yeruda pluton in the Transangarian segment (883 Ma (this study)) no magmatic events have been identified in the western framing of the Siberian craton.

In this study, we consider in more detail two additional lines of evidence. One of them is the structural-tectonic evidence related to the particularities and boundaries of the Central Angara terrane. The other is derived from paleomagnetic data. As we demonstrated above the Central Angara terrane is framed by the Priyenisei and Ishimba sutures, along which we find fragments of Precambrian ophiolite belts. This, along with the distinctive inner structure and deformation character different from that of the other terranes, reflects the "foreignness" of the Central Angara terrane. Moreover, because ophiolites are a most important indicator of geodynamic setting of rock formation and are fragments of ancient oceanic crust (Dobretsov and Zonenshain, 1985; Zonenshain et al., 1990), explaining how the Ribny–Panimba belt ophiolites formed within a coeval so called "Grenville-age collisional belt" would be very difficult indeed. It should also be noted that the supporters of "Grenville-age collisional events" on the Yenisei Ridge consider them on a small area and only within the Central Angara terrane, the maximum proportions of which do not exceed 400×80 km. Grenville-age collisional belts typically have transregional scales because they are related to continent–continent or continent–microcontinent collisions, and can be traced for several thousands of kilometers while being 200–400 km wide (Li et al., 2008; Rivers, 1997, 2008).

Our new paleomagnetic data obtained for the Yeruda pluton granitoids are a significant addition to the evidence already discussed. The paleomagnetic pole for the Central Angara terrane at the time of emplacement of the Yeruda granites is significantly different from the Siberian craton pole. Without delving into debatable questions and possible interpretation versions for the obtained data, and taking into account all available uncertainties and error parameters we can state that the distance between the Central Angara terrane and the Siberian craton was no less than a thousand kilometers at the time of the tectonic event that caused the emplacement of the Teya complex granites. Thus any conclusions concerning a "Grenville-age orogenesis" within the Priyenisei band of the Siberian continental margin involving the Central Angara terrane are unfounded.

The debate and uncertainty concerning stratigraphic correlations that advocate the unity of the Meso-Neoproterozoic sedimentary basins of the Central Angara and East Angara provinces (Kachevsky et al., 1998; Khomentovsky et al., 1972; Khomichev, 1999; Postelnikov, 1980) are also becoming clear. While the accumulation of sedimentary series of the East Angara terrane is absolutely related to the Siberian continental margin, the Central Angara block had its own independent terrane history and provenance areas.

In conclusion, detailed structural, petrological and geochemical, geochronological, and paleomagnetic investigations of the Yeruda pluton granites of the Teya complex belonging to the Central Angara terrane, as well as a general geological-geophysical analysis of its geology, has culminated in the solution of a very important issue of reconstructing the formation of the tectonic structure of the Yenisei Ridge and the western framing of the Siberian craton in the Meso-Neoproterozoic. This issue is essential, because its resolution allows us to introduce some constraints and perform geodynamic reconstructions for the Siberian craton and its Meso-Neoproterozoic framework much more accurately. Our integrated studies demonstrate that including the western framing of the Siberian craton in Grenville-age collisional events related to the formation of the Rodinia supercontinent is impossible.

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This is the contribution to the IGCP 648.

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